

FUNCTIONAL NITROXYLS FOR POLYPROPYLENE MODIFICATION

by

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Abstract

Chemistry for introducing a long-chain branch (LCB) architecture to linear, isotactic polypropylene (PP) is described, wherein degradation of the polymer matrix through β -scission of macroradicals is inhibited by an additive comprised of a nitroxyl that bears a polymerizable functional group. Trapping of alkyl macroradicals by the nitroxyl yields a macromonomer derivative that crosslinks when exposed to residual initiator-derived radicals. As such, formulations containing 4-acryloyloxy-2,2,6,6-tetramethylpiperidine-1-oxyl (AOTEMPO) and a peroxide transform linear PP into a gel-free, long-chain branched derivative (LCB-PP) that exhibits exceptional strain hardening with minimal loss in matrix molecular weight.

The chemistry is extended to produce PP bearing pendant anhydride and/or imide grafts. Formulations containing 4-vinyl ether-2,2,6,6-tetramethylpiperidine-1-oxyl (VETEMPO), maleic anhydride (MA), and peroxide produced anhydride-functionalized polypropylene (PP-*g*-MA). Following macroradical trapping by VETEMPO, pendant vinyl ether groups undergo a strictly alternating copolymerization with MA, thereby introducing anhydride without incurring large-scale matrix degradation. At a given melt viscosity, this technology provided PP-*g*-MA derivatives with higher graft content than standard peroxide-only reactions. Moreover, when combined with the difunctional monomer N,N'-1,3-phenylene-dimaleimide (PDM), VETEMPO enabled the production of highly crosslinked PP with an appreciable gel fraction.

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List of Abbreviations

alt – alternating copolymer

AOTEMPO – 4-acryloyloxy-2,2,6,6-tetramethylpiperidine-1-oxyl

b – branched polymer

BHT – butylated hydroxytoluene

bis-TEMPO – bis(1-oxyl-2,2,6,6-tetramethylpiperidine-4-yl)sebacate

BVE – butyl vinyl ether

°C – celsius

C=C – carbon-carbon double bond

CDCl₃ – deuterated chloroform

CI – carbonyl index

cm - centimeter

cm⁻¹ – wavenumber

¹³C NMR – carbon-13 nuclear magnetic resonance

CO₂ – carbon dioxide

d – doublet

DBTDL – dibutyl tin dilaurate

DCP – dicumyl peroxide

dd – doublet of doublets

DDSA – 2-dodecen-1-yl-succinic anhydride

DSC – differential scanning calorimetry

FN – functional nitroxyl

FTIR – Fourier transform infrared

g – grafted polymer

g – gram

G' – storage/elastic modulus

G'' – loss modulus

g_{PP} – gram polypropylene

h - hour

H – hydrogen

¹H NMR – proton nuclear magnetic resonance

HOTEMPO – 4-hydroxy-2,2,6,6-tetramethylpiperidine-1-oxyl

Hz – hertz

[IrCl(cod)]₂ – 1,5-cyclooctadiene-iridium(I) chloride dimer

J₁ – geminal coupling constant

J₂ – vicinal coupling constant

k – rate constant

kPa – kilopascal

L130 – 2,5-Bis(tert-butylperoxy)-2,5-dimethyl-3-hexyne

LCB – long-chain branching

lit – literature

m - meter

M – moles per litre

m – multiplet

MA – maleic anhydride

mg – milligram

MHZ – megahertz

min – minute

mL – millilitre

mm – millimeter

mmol – millimole

M_n – number-average molecular weight

mp – melting point

MW – molecular weight

M_w – weight-average molecular weight

N – newton

N_2 – nitrogen

NaCl – sodium chloride

NPM – N-phenyl maleimide

NRC – nitroxide radical coupling

Pa – pascal

PDM – N,N'-1,3-phenylene-dimaleimide (PDM)

PP – polypropylene

ppm – parts per million

rad - radian

s – second

s – singlet

S – strain hardening coefficient

sc – supercritical

t – triplet

TAM – triallyl trimesate

T_c – crystallization temperature

TEMPO – 2,2,6,6-tetramethylpiperidine-1-oxyl

TLC – thin layer chromatography

TMPTA – trimethylolpropane

TR – trapping ratio

tt – triplet of triplets

UV – ultraviolet

VBTEMPO - 4-vinylbenzoyloxy- 2,2,6,6-tetramethylpiperidine-1-oxyl

VETEMPO – 4-vinyl ether-2,2,6,6-tetramethylpiperidine-1-oxyl

VTES – vinyltriethoxysilane

W - watt

wt% - weight percent

δ – chemical shift

η – complex viscosity

η_0^+ – time-dependent elongational viscosity

η_E^+ – elongational viscosity

μL – microlitre

μm - micrometer

μmol - micromole

Chapter 1

Introduction

1.1 Effects of Long-Chain Branching on Polymer Rheology

A long-chain branch (LCB) polymer architecture has relatively few high-molecular weight branches that are covalently bound as pendant groups to a main chain. LCB is differentiated from short-chain branching by the branch length, with those described as ‘long’ being sufficient to form at least 2-3 entanglements [1]. Unlike vulcanized elastomers, chains in an LCB polymer are not crosslinked into a covalent network.

LCB architectures are well suited to polymer processing operations that impose extensional deformations, such as film blowing, thermoforming, and foaming [2] [3]. This is because in the melt state, LCB polymers exhibit extensional strain hardening, wherein stresses increase in a non-monotonic manner with increasing elongational strain. This is due to chain entanglement effects that relax more slowly than the typical timescale of typical deformations [4]. The linear PP materials produced by conventional polymerization processes lack this chain entanglement mechanism, necessitating the development of technology to convert them to LCB derivatives. Note that uniaxial and biaxial strain hardening are known to be correlated, [2] allowing uniaxial elongational experiments to be used in place of less convenient biaxial experiments even when the processing method of interest requires biaxial strain hardening.

The effects of LCB on polymer rheology also manifest in oscillatory shear rheology. LCB introduces slow relaxational modes to the relaxational spectrum, increasing the characteristic relaxation time [4]. As these modes are rather slow, LCB has a stronger effect on rheology as shear rate decreases; At higher frequencies LCB has little effect and shear thinning tends to dominate the rheological response. Therefore, at lower frequencies, viscosity (η) and storage modulus (G') of an LCB polymer are higher than a linear polymer of similar weight,

while at higher frequencies, molecular weight (MW) and dispersity will be the primary determinants of rheological behaviour.

Because rheology is more sensitive to long-chain branching than either of the alternative characterization methods (^{13}C NMR or triple detection gel permeation chromatography) [5], it serves as an excellent method of assessing the relative degree of branching across similar samples, and characterizing samples containing very low levels of branching. Additionally, rheological assessment relates directly to LCB's desired effect: increased processability.

1.2 Methods of Producing Long-Chain Branched Polypropylene

Polypropylene (PP) is an inexpensive, commercial polyolefin with good strength, crystallinity, and thermal stability. Given these desirable traits, there is naturally an interest in producing long-chain branched polypropylene (LCB-PP) which will be suitable for manufacturing processes requiring strain hardening.

Unfortunately, success in introducing LCB during the polymerization process has been quite limited for PP. Metallocene catalyst systems have only achieved very low branch densities [6]. Ideally, unsaturated macromonomers are formed either through catalyst activity or the incorporation of a difunctional comonomer, but steric limitations greatly hinder the incorporation of macromonomers to produce LCB. The mechanisms of polypropylene polymerization prove difficult obstacles to the polymerization of LCB-PP, in contrast with success achieved in producing LCB-polyethylene [6].

Branching and other graft modifications of polypropylene are commonly performed post-polymerization through radical-mediated chemistry, due to both the limitations of PP polymerization outlined above and the great accessibility and convenience of post-polymerization methods. Reactive extrusion is a particularly popular modification method, as processing the polymer as a melt allows for good mixing of additives without the use of a solvent. These

processes use a peroxide (or other radical source) to generate macroradicals through H-atom abstraction from the polymer. Tertiary carbon-centered radicals constitute a significant fraction of these macroradicals derived from PP [7], and their susceptibility to β -scission results in cleavage of polymer chains to reduce molecular weight [8].

Since melt-state reaction of PP with an organic peroxide results in large scale degradation of the polymer matrix, numerous attempts have been made to include multifunctional monomers such as triallyl trimesate (TAM) or trimethylolpropane triacrylate (TMPTA) to promote branching. These reagents possess multiple C=C groups that can graft through a radical addition/H-atom transfer sequence, leading to a branched polymer when multiple chains engage a given coagent molecule.

While these multifunctional monomers facilitate PP branching, they do not prevent β -scission from taking place concurrently [8]. Unfortunately, the low efficiency of monomer grafting to PP makes coagent-induced branching uncompetitive with chain scission, leading to substantially degraded matrix viscosities [9]. Moreover, the coagent grafting process is statistically skewed toward the larger polymer chain population, since higher molecular weight molecules have more H-atom donors. Since large chains have a greater tendency to branch, while degraded chains are statistically less likely to engage coagent, these reactions produce bimodal molecular weight and branching distributions [10]. These materials are typically comprised of hyper-branched nanoparticles suspended in a degraded matrix (Figure 1). This behaviour is intensified for branching agents capable of homopolymerization, such as triacrylates [9].

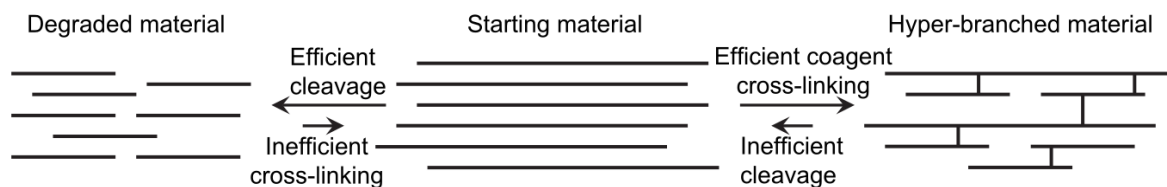


Figure 1 - Development of a bimodal weight distribution in LCB-PP (reproduced from [10])

One method of limiting the development of this bimodal distribution is to decouple the branching reaction from the crosslinking reaction. This typically involves a radical modification of PP, followed by non-radical crosslinking mechanism [11]. The two most common choices for grafted functionality are maleic anhydride (MA) and vinyltriethoxysilane (VTES). The former is generally crosslinked using di- or tri-functional coagents (such as diamines) that are reactive with the grafted anhydride group [12], while the latter relies on moisture curing of the silane group [13]. Beyond the β -scission incurred during grafting, these decoupled methods face several other downsides. Vinylsilane grafting is known to involve an intramolecular chain transfer of the radical to the polymer backbone following grafting, resulting in an uneven graft distribution with some chains receiving upwards of five or six grafts [14]. There is also some evidence that MA grafting has a similar tendency to introduce multiple grafts to single polymer chains [15]. Additionally, these methods require additional processing steps to carry out the crosslinking reaction after grafting. Addition of the crosslinking coagent concurrent with radical grafting will result in a branching process that behaves similarly to multifunctional monomer methods [16].

Operating below the melting point of PP can mitigate degradation by reducing the rate of scission. Solid-state modification employing ionizing radiation [17] or UV radiation in conjunction with a photoinitiator [18] can initiate coagent grafting, but efficiencies are limited by long exposure times and sample thicknesses. Recently *Diop et al.* [19] demonstrated the potential of solid-state shear pulverization to enable mixing below the polymer's melting point. Work by *Wang et al.* [20] suggests the presence of supercritical CO₂ in reactive extrusion may improve radical diffusion (accelerating branching and grafting reactions) while also serving as a foaming agent for the LCB reaction product.

1.3 Methods of Producing Maleic Anhydride-Grafted Polypropylene

While PP provides a wide range of valuable material properties at comparatively low cost, the low surface energy of this polyolefin compromises its interfacial adhesion when used in many filler-reinforced composites and polymer blend formulations. This deficiency can be rectified to some extent by grafting PP with MA to introduce a reactive, polar functional group [21]. However, as discussed above, radical grafting of MA is accompanied by extensive matrix degradation, and low MW PP-*g*-MA is generally blended with unmodified, higher MW PP to compensate.

The efficiency of MA grafting is compromised somewhat by the monomer's limited ability to oligomerize under melt modification conditions [22]. To achieve multiple anhydride groups on a PP chain, MA addition to backbone macroradicals must be accompanied by H-atom transfer from the polymer to a succinyl radical, leading to a close propagation sequence. Note that the same techniques developed to address β -scission in LCB-PP production have been applied to MA grafting, including solid-state shear pulverization [23] and scCO_2 co-extrusion [24]. Mediation of MA grafting by *N*-bromosuccinimide can yield high graft contents while preserving MW but racemizes the polymer, yielding a low-crystallinity polypropylene [25].

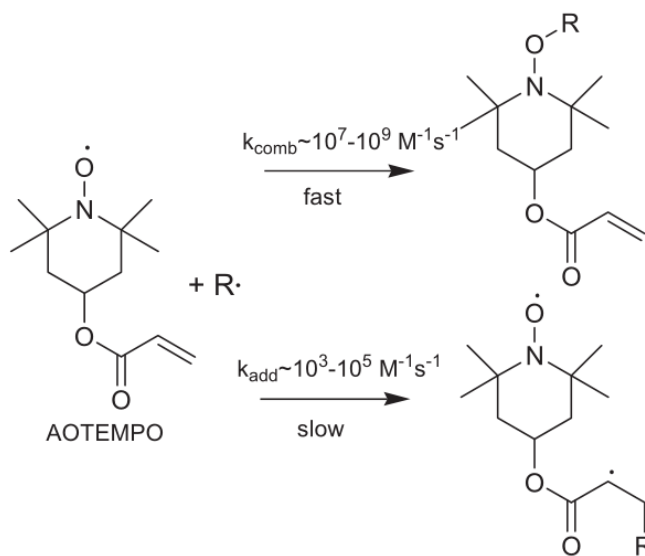
An approach unique to grafting MA is the use of styrene as a grafting coagent. Trapping of PP macroradicals with styrene produces pendant benzylic radicals that cannot fragment the polymer backbone. Perhaps more important is the ability of styrene and MA to form an alternating copolymer, producing styrene-MA copolymer sidechains [26]. Given styrene's volatility and toxicity, it is not an ideal coagent [21].

1.4 Radical-Mediated Modifications Using Functional Nitroxyl Chemistry

Nitroxyl radical coupling (NRC) reactions, and specifically the trapping of peroxide-generated macroradicals, are well-documented as a means of functionalizing scission-prone

polymers like PP [27]. TEMPO can serve as an effective scorch protectant for polyethylene crosslinking formulations, given that macroradical trapping by nitroxyl radicals operates close to the diffusion limit in the polymer melt ($k = 10^{-8}$ - $10^{-9} \text{ M}^{-1}\cdot\text{s}^{-1}$) [28]. This extremely quick quenching of macroradicals to produce spin-paired alkoxyamines inhibits other alternate macroradical chemistry [29], including grafting and β -scission. Functionalization of 4-hydroxy-TEMPO (HOTEMPO) opposite the nitroxyl radical does not disable its radical trapping efficiency, allowing the formation of the alkoxyamine to serve as a means of functionalizing the polymer chain while limiting macroradical lifetimes [30].

While there are no published studies on branching of PP using nitroxyls, variations of this chemistry have been used to produce crosslinked thermosets. One study [31] used silylated TEMPO-derivatives to produce moisture-curing derivatives. While not tested on PP, bis-TEMPO has been tested as a radical crosslinking agent for polyethylene [32]. Most importantly, previous work has demonstrated that TEMPO functionalized with a variety of oligomerizable C=C functionalities are capable of crosslinking polypropylene with limited matrix degradation [33]. The NRC reaction produces macromonomers that are oligomerized to crosslink the polymer after all nitroxyl is converted to alkoxyamine. This sequential trapping/oligomerization is a byproduct of a large difference in reaction rates, with radical oligomerization of a functional group such as an acrylate being orders of magnitude slower than radical coupling with the nitroxyl (Scheme 1) [29]. Because both reactions are radical-mediated and no additional chemicals must be added for the second reaction to proceed (assuming sufficient peroxide is present to carry out both reactions), the entire modification can be carried out in a single process.



Scheme 1 - Alkyl radical reactions with acrylated TEMPO (AOTEMPO) (reproduced from [34])

Identifying the correct ratio of peroxide to functional nitroxyl is crucial to efficient modification. The term ‘trapping ratio’ (TR) [29] was coined to describe the relative amounts of nitroxyl and peroxide-derived radicals, and is defined as:

$$TR = \frac{[\text{nitroxyl}]}{\text{radicals per peroxide molecule} \times [\text{peroxide}]}$$

If the trapping ratio is too high, insufficient radicals remain after the nitroxyl conversion to support macromonomer oligomerization. If the trapping ratio is too low, uninhibited radical chemistry will begin to dictate product properties after the macromonomer is consumed. This is of great concern in PP, where cure reversion will occur if β -scission outpaces oligomerization [33].

An additional restriction on functional nitroxyl modification is that not all trapped radicals will be macroradicals. While nitroxyl radicals will not trap the alkoxy radicals produced

by thermolysis of peroxides, fragmentation of the alkoxy radicals to ketone + $\text{CH}_3\cdot$ will be followed by trapping of the methyl radical, yielding a methylalkoxyamine [7]. The rate of alkoxy fragmentation and ease of H-atom abstraction from the polymer dictate the macroradical yield and, by extension, the yield of the polymer-bound alkoxyamine.

Dicumyl peroxide (DCP) has an abstraction efficiency to PP of 0.37 at 160°C [7], meaning only 37% of cumyloxy radicals abstract an H-atom from PP, with the remainder fragmenting to methyl radicals. One advantage provided by functional nitroxyl chemistry is its ability to incorporate methyl alkoxyamines into the polymer network by co-oligomerization with macromonomer functionality.

One potential complication of nitroxyl-mediated PP modifications is the instability of tertiary alkoxyamines toward disproportionation to olefin + hydroxylamine [32]. In a model compound study by *Garret et al.* [7] DCP favoured H-atom abstraction from a PP analogue (2,4-dimethylpentate) forming tertiary alkyl radicals over primary or secondary radicals by an 11:6 ratio. Inhibited H-atom abstraction from the secondary position was attributed to steric hinderance [35]. It is therefore clear that tertiary alkoxyamines are formed when functionalizing PP using TEMPO-based reagents, but the impact of their instability on reaction yields and product properties is not well understood [27].

1.5 Fundamentals of Charge-Transfer Alternating Copolymerization

The aforementioned copolymerization of MA and styrene is an example of ‘charge-transfer’ polymerization, in which an electron-rich and an electron-poor monomer form an alternating copolymer. While the mechanism of this radical copolymerization is unclear [36] [37], it is known that partial charge transfer between the donor and acceptor monomers plays a role in the preference for alternating monomer addition. If homopolymerization of either monomer is disfavoured, the resulting copolymer will be strictly alternating.

Maleic anhydride is commonly used as an acceptor monomer in these polymerizations, and it has been shown to copolymerize in a strictly alternating manner with vinyl ethers [38]. In fact, MA-*alt*-vinyl ether copolymers have been the subject of much research, and are commercially available [39] [40] [41]. Aside from the work discussed previously using styrene as a grafting coagent for MA, little work has been done investigating the behaviour or potential of these reactions for melt-state modification at low monomer concentrations.

1.6 Research Objectives

Given PP's tendency to degrade during radical-mediated modification, functional nitroxyl chemistry is a promising pathway for performing a variety of structural and functional transformations. In Chapter 3, previous research on PP thermoset production with functional nitroxyls [33] is scaled back to introduce LCB to linear PP using acrylated TEMPO (AOTEMPO). The goal is to produce LCB-PP with a uniform branch distribution without incurring losses in molecular weight, something that is currently impossible by melt-state polymer modification chemistry. The efficacy of the branching method and its effects of polymer architecture will be characterized using melt rheology and differential scanning calorimetry.

Chapter 4 describes an exploration of alternating copolymerization as a means of maleating PP without incurring large-scale matrix degradation. Vinyl ether functionality will be introduced to PP by the NRC grafting of vinyl ether-functionalized TEMPO (VETEMPO). Following radical trapping, the pendant vinyl ether functionality will copolymerize with free MA to incorporate anhydride functionality into the polymer matrix. The success of this approach will be gauged by measurements of melt-state rheological properties and polymer-bound anhydride functional group concentration.

Chapter 2

Experimental Methods

2.1 Materials

All chemicals were used as received unless otherwise stated. Isotactic polypropylene (PP, $M_w = 340\,000$ g/mol, $M_n = 97\,000$ g/mol), dicumyl peroxide (DCP, 99%, 2,2,6,6-tetramethylpiperdin-1-oxyl (TEMPO, 98%), 4-hydroxy-2,2,6,6-tetramethylpiperdin-1-oxyl (HOTEMPO, 97%), acryloyl chloride (>97%), 1,5-cyclooctadiene-iridium(I) chloride dimer ($[\text{IrCl}(\text{cod})]_2$, 97%), vinyl acetate (>99%), sodium carbonate (>99%), 2-dodecen-1-yl-succinic anhydride (DDSA, 95%), pyridine (>99.9%), ethylene chloroformate (97%), benzylamine (99%), vinyl triethoxysilane (VTES, 97%), butylated hydroxytoluene (BHT, >99%), maleic anhydride (>99%), butyl vinyl ether (BVE, 98%), naphthalene (>99%), N-phenylmaleimide (NPM, 97%), and N,N'-1,3-phenylene-dimaleimide (PDM, 97%) were purchased from Sigma-Aldrich. 2,5-bis(tert-butylperoxy)-2,5-dimethyl-3-hexyne (L-130, 56%) was purchased from Elf Atochem, while dibutyl tin dilaurate (DBTDL, >94%) was purchased from Alfa Aesar. DDSA was chemically dehydrated (as described below) to reverse partial hydrolysis of the anhydride.

2.2 Synthesis of TEMPO Derivatives

2.2.1 AOTEMPO

AOTEMPO was prepared as described previously [42], with minor modifications. A solution of triethylamine (4.30 mL, 3.12 g, 30.9 mmol) in dry benzene (14 mL) was added dropwise to a stirred solution of HOTEMPO (1.36 g, 7.90 mmol) in dry benzene (20 mL) at room temperature under N_2 . A solution of acryloyl chloride (675 μL , 0.379 g, 4.18 mmol) in dry benzene (14 mL) was added dropwise at room temperature, still under N_2 with stirring. The

solution was stirred for 24 hours, after which a second dropwise addition of acryloyl chloride (340 μ L, 0.379 g, 4.18 mmol) in benzene (7 mL) was performed. The reaction was stirred an additional hour, while monitoring reaction progress through TLC. The solution was vacuum filtered twice, washing with benzene. Solvent was removed from the filtrate under vacuum, yielding orange crystals. These crystals were recrystallized from cyclohexane to yield the purified product, which was characterized by melting point. Yield: 76%; mp 96-97°C; lit 102-104 °C [42].

Since the paramagnetic nature of the radical prevents effective NMR characterization, AOTEMPO was reduced to corresponding *N*-hydroxyl amine using phenylhydrazine. ¹H-NMR (400 MHz, CDCl₃, δ , ppm): 6.29 (1H, dd, $J_1 = 2$ Hz, $J_{2(\text{trans})} = 17$ Hz), 5.99 (1H, dd, $J_{2(\text{trans})} = 17$ Hz, $J_{2(\text{cis})} = 11$ Hz), 5.72 (1H, dd, $J_1 = 2$ Hz, $J_{2(\text{cis})} = 11$ Hz), 5.05 (1H, tt, $J_{2(\text{trans})} = 11$ Hz, $J_{2(\text{cis})} = 4$ Hz), 1.89 (2H, dd, $J_1 = 12$ Hz, $J_{2(\text{cis})} = 4$ Hz), 1.65 (2H, t(broad), $J_1 = 12$ Hz, $J_{2(\text{trans})} = 11$ Hz), 1.19 (6H, s), 1.16 (6H, s)

2.2.2 VETEMPO

VETEMPO was prepared as described previously [43], with minor modifications. A solution of HOTEMPO (2.51 g, 14.6 mmol), sodium carbonate (1.538 g, 14.5 mmol), and [IrCl(cod)]₂ (0.195 g, 0.290 mmol) in dry toluene (30 mL) was heated under N₂ to 90 °C while stirring. Vinyl acetate (2.68 mL, 2.50 g, 29.0 mmol) was added and the mixture stirred for 5 hours, at which point the solution was removed from heat and allowed to stir overnight. The solution was concentrated by rotary evaporation and added to hexanes (50 mL) and stirred for 70 minutes before vacuum filtering and washing with hexanes. The filtrate was concentrated by rotary evaporation and purified by column chromatography using a mobile phase of 5:1 hexanes/ethyl acetate. Solvent removal by rotary evaporation yielded a dark red liquid, which when placed in a freezer overnight, provided orange-red crystals. Yield: 24%; mp 30-31°C

Since the paramagnetic nature of the radical prevents effective NMR characterization, VETEMPO was reduced to the corresponding *N*-hydroxyl amine using phenylhydrazine. ¹H-NMR (400 MHz, CDCl₃, δ , ppm): 6.21 (1H, dd, $J_{2(\text{cis})} = 7$ Hz, $J_{2(\text{trans})} = 14$ Hz), 4.19 (1H, dd, $J_1 = 2$ Hz, $J_{2(\text{trans})} = 14$ Hz), 4.01 (1H, tt(broad)), 3.94 (1H, d(broad), $J_1 = 2$ Hz, $J_{2(\text{cis})} = 7$ Hz), 1.92 (2H, d(broad), $J_1 = 12$ Hz), 1.58 (2H, m (broad)), 1.19 (6H, s), 1.13 (6H, s). See Appendix A for peak assignment.

2.3 Synthesis of Butyl Vinyl Ether-*alt*-Maleic Anhydride

Dried tetradecane (8 mL, 6.1 g), maleic anhydride (62.1 mg, 0.633 mmol), naphthalene (4.6 mg, 0.036 mmol), and dicumyl peroxide (4.2 mg, 0.016 mmol) were mixed in a glass vial and degassed with N₂. Butyl vinyl ether (80 μ L, 62.4 mg, 0.623 mmol) was added by syringe. The solution was heated under stirring to 80 °C and shaken to mix thoroughly once MA had melted. The solution was heated for 10 minutes, after which it was allowed to cool to room temperature. The precipitate was collected by vacuum filtration and washed with cold hexanes, before drying under vacuum. Yield: 31%

2.4 Dehydration of Partially Hydrolyzed Dodecenyl Succinic Anhydride

Having partially hydrolyzed to dodecenyl succinic acid in storage, dodecenyl succinic anhydride (DDSA) had to be dehydrated to ensure high purity of the anhydride functionality. In a 250 mL round bottom flask fitted with a reflux condenser, partially hydrolyzed dodecenyl succinic anhydride (2.285 g, 8.58 mmol) was dissolved in toluene (50 mL) with pyridine (0.5 mL, 0.491 g, 6.21 mmol). Under reflux and N₂, a mixture of ethyl chloroformate (1 mL, 1.14 g, 10.5 mmol) and toluene (10 mL) was added, and the flask was stirred for 30 minutes. The solution was vacuum filtered to remove the precipitate, then solvent was removed under vacuum. The crude product was purified by column chromatography using dichloromethane as eluent. Solvent was removed from isolated fractions by rotary vacuum, and the brown liquid yielded was further dried

under vacuum at 60°C for 24 hours. Placing the product in the freezer yielded a waxy, white-yellow solid. Yield: 47%; mp 40-41 °C; lit 41-43 °C [44].

2.5 Polypropylene Derivatization

2.5.1 General Compounding Procedure

Polypropylene (5 g) was ground and divided evenly between three 60 mL plastic bottles. Desired reagent loadings for the formulation were measured and dissolved in acetone (1.5 mL). The solution was then distributed evenly between the polymer powder within the bottles, and tumble mixed intermittently until the acetone had evaporated (2-3 hours).

An Atlas Laboratory Mixing Molder heated to 180°C was charged with the contents of one bottle and the compound was mixed for 18 minutes (5 half-lives of the peroxide). The product was then removed from the mixer, and the process was repeated for each subsequent bottle. Products from different bottles in the same batch were ground and combined.

2.5.2 PP-*b*-VTES

Peroxide-initiated graft modification of PP with vinyltriethoxysilane was carried out as described in Section 2.5.1. The combined PP-*g*-VTES samples were subsequently solution-coated with 5 µL/g DBTDL and 5 wt% pentaerythritol tetrakis(3,5-di-*tert*-butyl-4-hydroxyhydrocinnamate) and mixed using the Atlas Laboratory Mixing Molder for 3 minutes. The resulting stabilized PP-*g*-VTES samples were pressed into films approximately 1 mm thick using a Wabash press, then moisture-cured by submerging in boiling water for 2 hours.

2.6 Instrumentation and Analysis

2.6.1 Rheological Characterization

Compression molded discs 25 mm in diameter and >1 mm in thickness and sheets 0.5-0.7 mm in thickness were prepared in a Wabash hydraulic press for shear and extensional rheological

characterization, respectively. Storage modulus (G'), loss modulus (G''), and complex viscosity (η) were measured using an Anton Paar MCR-301 rheometer equipped with a CTD 180 air convection oven. The 25 mm diameter parallel plates were applied using a gap of 1 mm. Samples were measured over a frequency range of 0.05 rad/s to 100 rad/s at a constant strain of 3% and a temperature of 180 °C. Strain sweeps were used to confirm that measurements were acquired in the polymer's linear viscoelastic region. Time sweeps demonstrated a measure of instability for PP-*b*-AOTEMPO samples, with degradation taking effect after around 10 minutes. This limited the number of data points acquired and the range of frequencies that could be examined in shear rheology.

Uniaxial extensional rheometry was performed using the same rheometer and oven using a universal testing platform (Anto Paar SER-2). Samples were cut to a width of 10 mm, and testing was performed at 180 °C and strain rates of 0.1, 1, and 10 s⁻¹.

2.6.2 Gel Content

Gel content was determined in accordance with the ASTM D2765-16 test method [45]. Weighed samples were enclosed in 120 mesh stainless steel cloth and extracted in xylenes (with 1 wt% BHT to limit polymer degradation) under reflux for 12 hours. Samples were then dried to constant weight, and gel content reported as the mass fraction remaining of the original sample.

2.6.3 Differential Scanning Calorimetry

Differential Scanning Calorimetry (DSC) was performed using a TA Instruments DSC Q100. Samples were tested in hermetically sealed aluminum pans. All samples were held for 3 minutes at 200 °C to remove any thermal history of the samples prior to measurement. For nonisothermal testing, samples were cooled from 200 °C to -40 °C at a rate of 5 °C/min, then held at -40 °C for 5 minutes before heating the sample back to 200 °C at the same rate. For isothermal

testing, samples were cooled to their target temperature (130 or 140 °C) at a rate of 30 °C/min before maintaining the target for 1 h.

2.6.4 Hot Stage Microscopy

Isothermal crystallization experiments were performed using a Linkam CSS 450 hot stage mounted on an Olympus BX51 optical microscope. Samples were heated to 200 °C and pressed into a thin film by reducing the plate gap to 50 μm , and the temperature was held at 200 °C for 5 minutes to eliminate the sample's thermal history. The film was then cooled to 140°C at a rate of 30 °C/minute and the temperature was kept constant for 30 minutes. Images were captured using a Sony ExwaveHAD 3 CCD digital recorder.

2.6.5 Transmission FTIR

Fourier transform infrared (FTIR) spectrometry was performed using a Bruker Alpha II spectrometer equipped with a universal sampling (transmission) module. Thin films were prepared by either pressing the sample using a Wabash hydraulic press or (for BVE-*alt*-MA) by casting a film on an NaCl disk. Film casting was accomplished by dissolving the sample in acetone, transferring the solution to the surface of a [25 mm diameter, 4 mm thickness] NaCl disk, evaporating the acetone in a vacuum oven at 60 °C. Sample spectra were acquired in triplicate from 64 scans at a resolution of 2 cm^{-1} .

2.6.6 Graft Content Determination

Modified PP samples for FTIR analysis were purified by dissolving the material (0.25 g) in xylenes (15 mL) under reflux with 0.5 g BHT, then precipitating in methanol (80 mL) and drying under vacuum. FTIR spectroscopy was performed using the procedure outlined above. Anhydride content was determined by indexing the area of the carbonyl peak (1760-1812 cm^{-1}) against an internal standard peak arising from the base resin (2695-2753 cm^{-1}). Carbonyl index (CI) was calculated as:

$$CI = \frac{A_{1760-1812}}{A_{2695-2853}}$$

The concentrations of single anhydride grafts and grafted anhydride-*alt*-vinyl ether functionality in purified materials were quantified by appropriate FTIR calibrations. Calibration standards were prepared by blending a representative compound (DDSA for the single graft, BVE-*alt*-MA for the grafted copolymer) with polypropylene using the general compounding procedure described above, substituting a 3-minute compounding time.

Graft degree is defined as the weight ratio of the grafted MA to the bulk material. Graft efficiency is defined as the ratio of grafted MA to the MA initially added.

2.6.7 Reaction Dynamics

The progress of select PP modifications was modified continuously by conducting reactions in the melt-sealed cavity of a control-strain rheometer (Alpha Technologies Advanced Polymer Analyzer 2000) equipped with biconical plates operating at 180 °C with a shear arc and frequency of 0.5° and 1 Hz, respectively. Formulations were prepared by coating ground polymer (4 g) in an acetone solution of the desired reagents and allowing the acetone to evaporate (1 hour), tumble mixing intermittently.

Chapter 3

AOTEMPO-Mediated Branching of Polypropylene

3.1 Introduction

Polypropylene (PP) is a commodity polyolefin that is valued for its strength, crystallinity, and thermal stability. Unfortunately, catalyst limitations restrict polypropylene polymerization to linear chains [6] [46] that are ill-suited to many melt state polymer processing operations. Introducing long-chain branching (LCB) improves extensional melt viscosity by inducing strain hardening characteristics, thereby allowing the material to be used in thermoforming, foaming, and film blowing applications [2].

LCB-PP can be prepared by chemical modification of linear parent materials using radical chemistry to accomplish grafting and/or crosslinking reactions. Unfortunately, macroradical intermediates generated by H-atom abstraction from the polymer are susceptible to β -scission, resulting in chain cleavage and degradation of the polymer matrix [8]. In order to increase extensional viscosity effectively, chain branching modifications must not only introduce long-chain branches, but also build sufficient molecular weight to overcome the weight losses incurred from β -scission.

Strategies to mitigate β -scission include operating at temperatures below the melting point of PP, as practiced for solid-state shear pulverization [19] and irradiation [18] technologies. In the case of melt-state modifications, however, the general approach is to offset chain scission by inducing a significant degree of chain crosslinking. Multifunctional monomer coagents such as triallyl trimesate (TAM) and trimethylolpropane triacrylate (TMPTA) have been applied to LCB-PP production, but the degree of modification that is needed to counteract β -scission results in a bimodal molecular weight distribution that is comprised of a small population of hyper-branched chains dispersed in a matrix of degraded linear chains [10]. Two-step branching methods narrow

this bimodal distribution by combining the radical grafting procedure with a subsequent non-radical crosslinking mechanism [11]. Examples include grafting of vinyl triethoxysilane (VTES) and moisture-curing the PP derivative to induce crosslinking, and maleic anhydride grafting [12] followed by crosslinking through the reaction of di- or tri-functional amines with the pendant anhydride functionality.

Previous research has shown functionalized nitroxyls to be suitable additives for the preparation of polypropylene thermosets [33]. Grafting is achieved through a nitroxyl radical coupling (NRC), in which the additive grafts by trapping macroradicals. Since this reaction operates close to the diffusion limit in polymer melt ($k = 10^8$ - $10^9 \text{ M}^{-1}\cdot\text{s}^{-1}$) [28], macroradical lifetimes are suppressed to the point where polymer degradation is extinguished. Once the radical trapping functionality is exhausted, initiator-derived radicals then proceed to oligomerize the polymer-bound C=C functionality, covalently bonding chains together [29].

In this chapter, the ability of acrylated TEMPO (AOTEMPO) to support single-pot preparation of gel-free LCB-PP is demonstrated along with evidence of remarkable strain hardening characteristics and matrix viscosity.

3.2 Results and Discussion

3.2.1 Trapping Ratio Screening

As has been outlined in previous research [47] [34], radical-mediated modifications involving functionalized nitroxyls progress through three distinct phases: (1) An induction stage where peroxide-derived alkyl radicals are trapped by nitroxyls to give alkoxyamines, (2) an oligomerization stage where oligomerization of polymer-bound acrylate functionality competes with other radical chemistry, and (3) a stage where conventional polymer modifications occur unencumbered. In the case of PP, stage 3 involves degradation arising from β -scission and is referred to as ‘cure reversion’ [33]. The duration and yield of each phase is dictated by the

initiator decomposition rate as well as the relative quantities of peroxide and nitroxyl. *Hyslop et al.* [29] have described this relationship in terms of the ‘trapping ratio’ (TR), defined by

$$TR = \frac{[nitroxyl]}{radicals\ per\ peroxide\ molecule \times [peroxide]}$$

The effects of LCB on oscillatory shear rheology are known to decrease with increasing shear rate, as high frequency dynamic responses are influenced by small segment dynamics [4]. For this reason, storage modulus (G') and viscosity (η) data acquired at $100\ s^{-1}$, the high end of the frequency range tested, can be used for qualitative comparisons of matrix molecular weight. At lower frequencies, chain entanglement effects generated by LCB result in higher G' and η values compared to linear polymers of the same weight. As such, data acquired below 1 Hz are sensitive to the LCB chain population of a given sample. Loss modulus (G'') relatively insensitive to LCB effects [4].

Preliminary experiments were conducted to identify suitable formulations for LCB-PP preparation. These trials used a fixed peroxide loading along with varying trapping ratios achieved by adjusting the concentration of AOTEMPO. The oscillatory shear rheology of the products is illustrated in Figure 2.

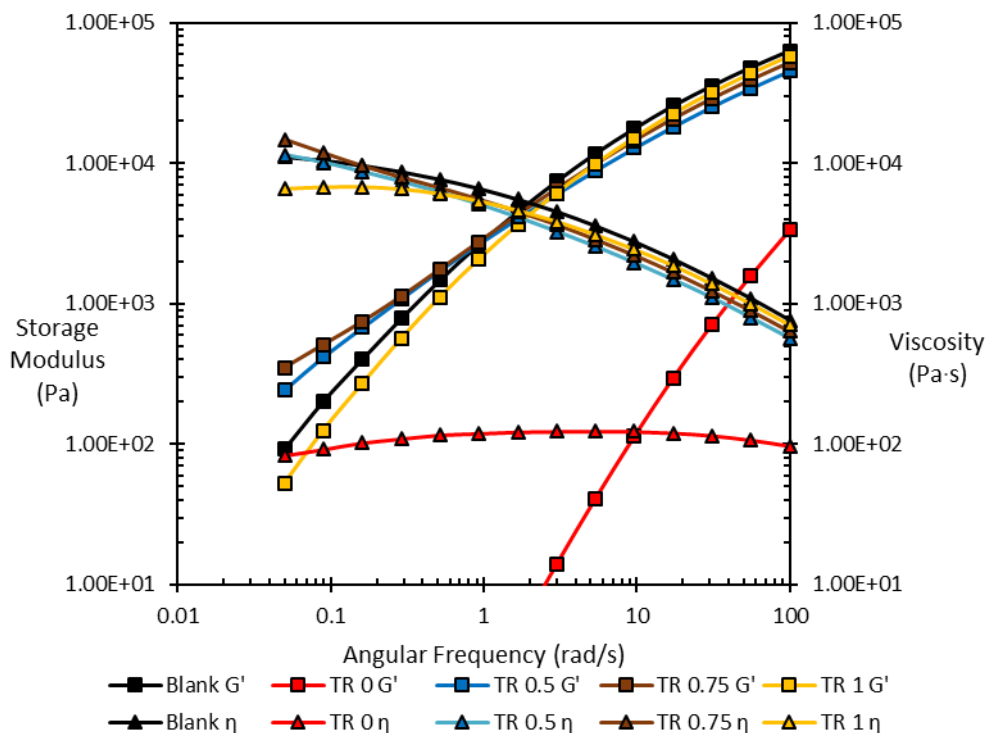


Figure 2 - Viscosity and storage modulus of modified PP, varying TR while [L130] remains fixed at 8 $\mu\text{mol/g}$

As expected, PP degraded significantly when heated with peroxide in the absence of the functional nitroxyl (TR = 0), as evidenced by a loss of both complex viscosity and storage modulus. Including AOTEMPO in the formulation inhibited β -scission by quenching macroradicals to form the polymer-bound alkoxyamine, thereby preserving melt viscosity during stage one of the process [33]. Additionally, diversion of the radical population during stage 2 away from PP macroradicals and toward oligomerization of the nitroxyl-grafted acrylate functionality may mitigate β -scission during this phase of the process. The ability of AOTEMPO to maintain the matrix molecular weight of LCB-PP is evident in the high-frequency η and G' of nitroxyl-based formulations, which are close to those of the original material.

With regards to branching, the expected rise in low-frequency η and G' [4] was observed in samples produced with trapping ratios of 0.5 and 0.75. Branching was not expected at a TR of

1, given that all initiator-derived radicals are expected trapped by the nitroxyl functionality, leaving none to carry out the chain-branching oligomerization. Nevertheless, some degradation of the material was observed, likely due to the thermal instability of polypropylene. A trapping ratio of 0.75 yielded higher η and G' than a trapping ratio of 0.5, indicating that cure reversion can occur when the formulation resides in stage 3 for excessive periods due to application of insufficient nitroxyl/excessive peroxide [47].

A second trapping ratio study was performed, wherein trapping ratio was varied by changing the peroxide loading at a fixed AOTEMPO concentration. This study produced similar rheological outcomes (Figure 3) to the previous TR experiments, albeit with several subtle differences. As with the previous study, a TR of 0.75 produced a polymer with similar high-frequency properties to the original material, along with increased low-frequency η & G' characteristic of LCB.

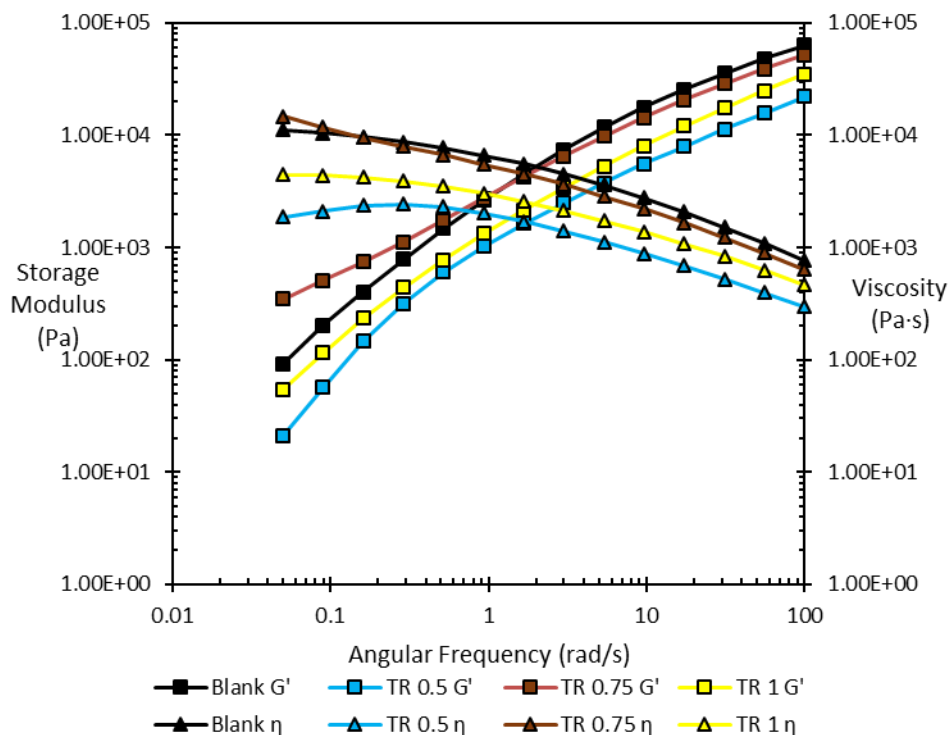


Figure 3 - Viscosity and storage modulus of modified PP, varying TR while [AOTEMPO] remains fixed at 24 $\mu\text{mol/g}$

In this study, however, a TR of 0.5 yield a much less viscous material than in the fixed-peroxide study. This is because the total loading here is lower: trapping ratio was achieved by increasing the peroxide loading rather than reducing the nitroxyl loading (compared to the TR 0.75 sample, which is identical in both), resulting in more degradation of the polymer network.

In previous studies of PP thermosets, the time evolution of G' was used to identify the transition between various reaction stages. For example, the transition from phase 2 to phase 3 of an AOTEMPO-based PP modification is identified by the point where the storage modulus reaches an upper plateau momentarily, before the decline associated with stage 3. In the present study, samples produced by a fixed-nitroxyl trapping ratio can be interpreted as ‘snapshots’ of a

single theoretical cure, with decreasing trapping ratio (increased peroxide loading) yielding a product further into this process. A trapping ratio of 1 places the melt at the boundary of the first and second stages, where the nitroxyl functionality has been entirely consumed but oligomerization has yet to occur extensively. Trapping ratios of 0.5 and 0.75 yield products somewhere in the range of the second and third stages. A TR of 0.5 can confidently be placed in the cure reversion stage, as indicated by lower G' and η values relative to the TR 0.75 formulation. Without time-resolved rheology or further sampling, the exact position of the TR 0.75 sample with respect to the oligomerization/reversion crossover cannot be determined. Nevertheless, evidence of MW preservation and branching in the low-frequency shear rheology established a TR of 0.75 as suitable for further investigation.

3.2.2 Polypropylene Branching Study

In order to demonstrate the advantages of AOTEMPO as a branching agent, its capabilities needed to be compared against those of an existing method. Of previously documented branching methods, moisture-cured PP-*g*-VTES was chosen as reference, due to its reduced tendency to produce bimodal MWD [11] compared to multifunctional monomer methods. For the purposes of processability, crosslinking to the gel point such that any fraction of the polymer has virtually infinite molar mass is to be avoided. Therefore, it is important that at least one sample prepared by AOTEMPO and silane chemistry be free of an insoluble gel fraction.

Five compounds were prepared to study the effectiveness of AOTEMPO as a radical-mediated PP branching agent (Table 1). As discussed above, all AOTEMPO preparations used a TR 0.75 and differed only in terms of the peroxide loading. The highest degree of PP modification, brought on by a relatively high initiator concentration (PP-AO3) produced an isolable amount of gel, thereby establishing a ceiling on the degree of modification possible while

remaining a fully soluble PP-LCB derivative. Likewise, a measurable gel fraction was observed when a high VTES concentration was employed (PP-S2).

Table 1 - Recipes used to prepare LCB-PP

Sample	[L130] ($\mu\text{mol/g}_{\text{PP}}$)	[AOTEMPO] ($\mu\text{mol/g}_{\text{PP}}$)	[VTES] ($\mu\text{mol/g}_{\text{PP}}$)	Gel Fraction (xylenes insoluble)	X_c
PP-AO1	6.0	18	–	<0.5%	43%
PP-AO2	8.0	24	–	<0.5%	44%
PP-AO3	9.7	29	–	5%	43%
PP-S1	1.5	–	263 (3 wt%)	<0.5%	46%
PP-S2	1.5	–	350 (4 wt%)	2%	46%

At these loadings

3.2.3 Oscillatory Shear Rheometry

As discussed in Section 3.2.1, oscillatory shear rheology provides insight into the architecture of the LCB chain population as well as the overall molecular weight of the PP derivative. Of particular interest are η and G' comparison between PP-*b*-AOTEMPO and PP-*b*-VTES derivatives at high shear frequencies (which increase with molecular weight) and low shear frequencies (where LCB effects η and G').

Analysis of the silane-modified samples (Figure 4) reveal significant decreases in both η & G' due to degradation of the polymer matrix. This is characteristic of radical-mediated PP modifications, as radical grafting occurs contemporaneously with β -scission [8]. In PP-S2 the effects of the crosslinked fraction are visible in low-frequency rheology, with G' being close to that of the untreated material at 0.567 rad/s, despite the branched sample's modulus being much

lower at high frequencies. This pattern of high-frequency losses from matrix degradation alongside low-frequency gains from crosslinking is typical of radical-mediated branching processes [11].

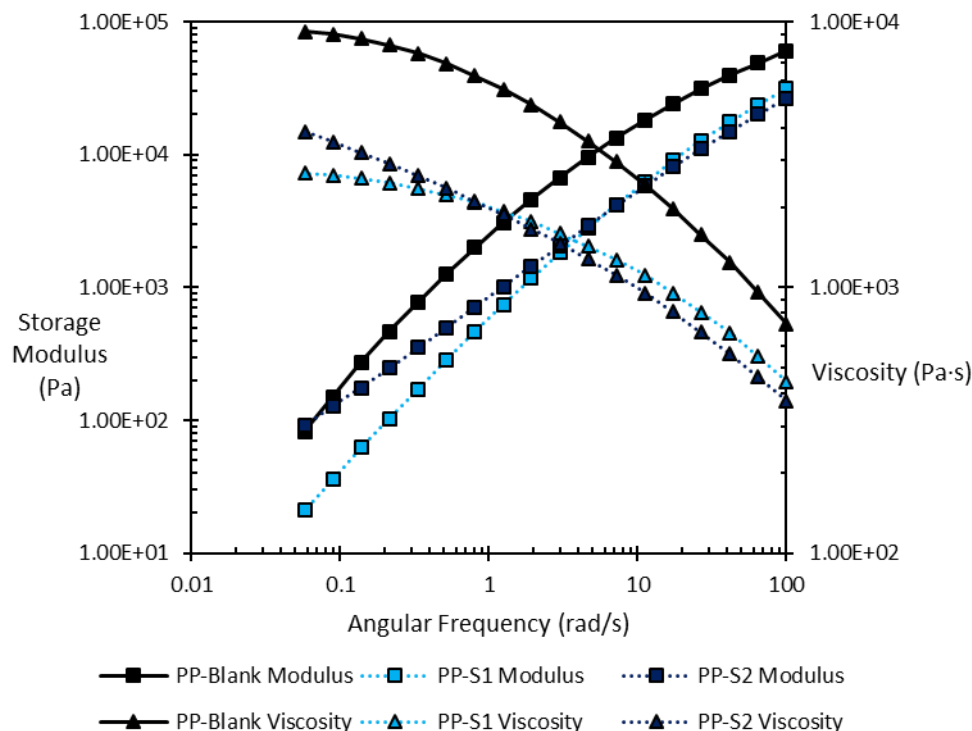


Figure 4 – Oscillatory shear rheology of VTES-branched samples

On account of its lower silane loading, PP-S1 was less functionalized and so less branched [48], resulting in lower η & G' at low frequencies than PP-S2, despite incurring similar matrix degradation, as evidenced by losses in high-frequency η & G' .

PP-*b*-AOTEMPO derivatives exhibited superior matrix integrity compared to PP-*b*-VTES (Figure 5). All samples showed minimal degradation, with moduli and viscosities at 100 Hz close to those of the unmodified starting material. Sample PP-AO1 shows that insufficient amounts of AOTEMPO and peroxide fail to produce a perceivable degree of LCB. PP-AO3 is at the opposite end of the modification spectrum, with a substantial gel fraction that eliminates the possibility of

a low-frequency Newtonian plateau for η and elevates G' to approach an equilibrium thermoset modulus [4].

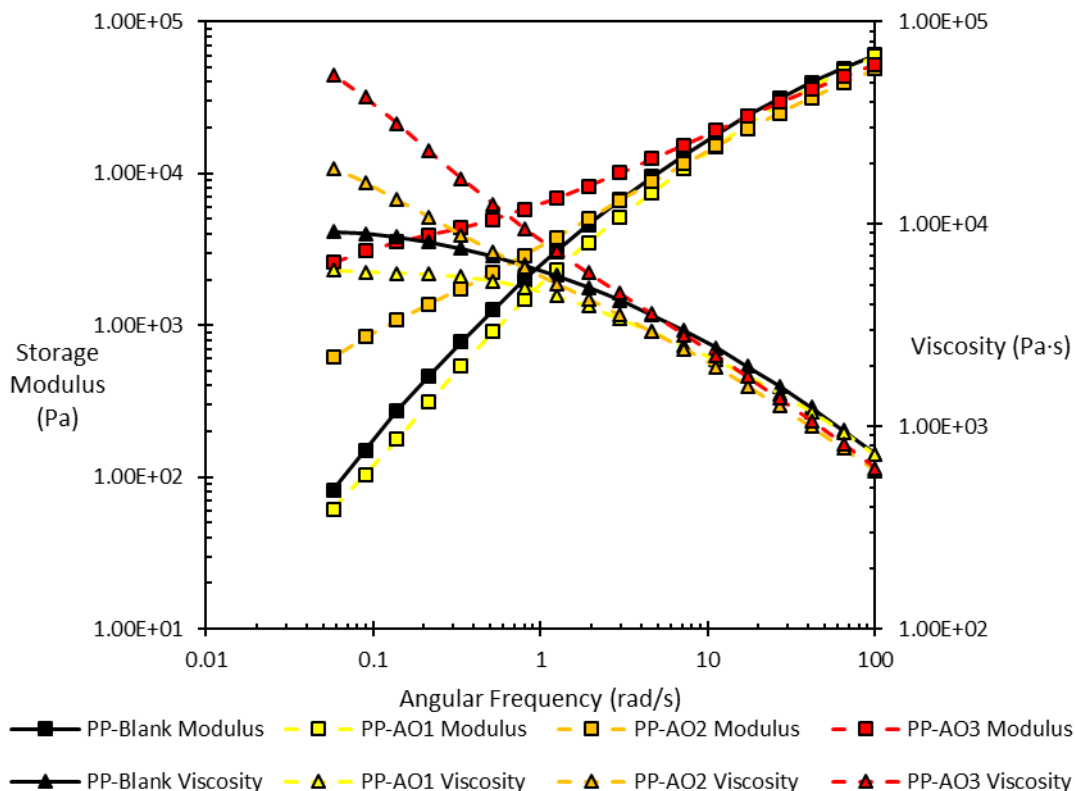


Figure 5 – Oscillatory shear rheology of AOTEMPO-branched samples

PP-AO2 is the ideal embodiment of LCB-PP achievable with AOTEMPO chemistry, as this formulation generated unambiguous evidence of long-chain branching while preserving the polymer's molecular weight. Since there was no crosslinked gel in this sample, a Newtonian viscosity plateau might be expected at very low oscillation frequencies. However, limitations on the shear rates that can be accessed for this material in this measurement configuration precluded data acquisition in this frequency domain.

Changes in loss angle over shear frequency further categorize PP-AO2 as an ideal LCB-PP (Figure 6). A plateau in loss angle is observed in PP-AO2 around 0.1-1 Hz, characteristic of the time-dependent relaxation characteristics of the long-chain branched material [49]. Gelled samples show strong retention of elasticity at low frequencies, while less-modified samples PP-AO1 and PP-S1 exhibit wholly convex curves similar to the linear base material. Additionally, at high frequencies, silane-branched samples are less elastic in their behaviour, owing to the matrix degradation experienced during peroxide-initiated graft modification.

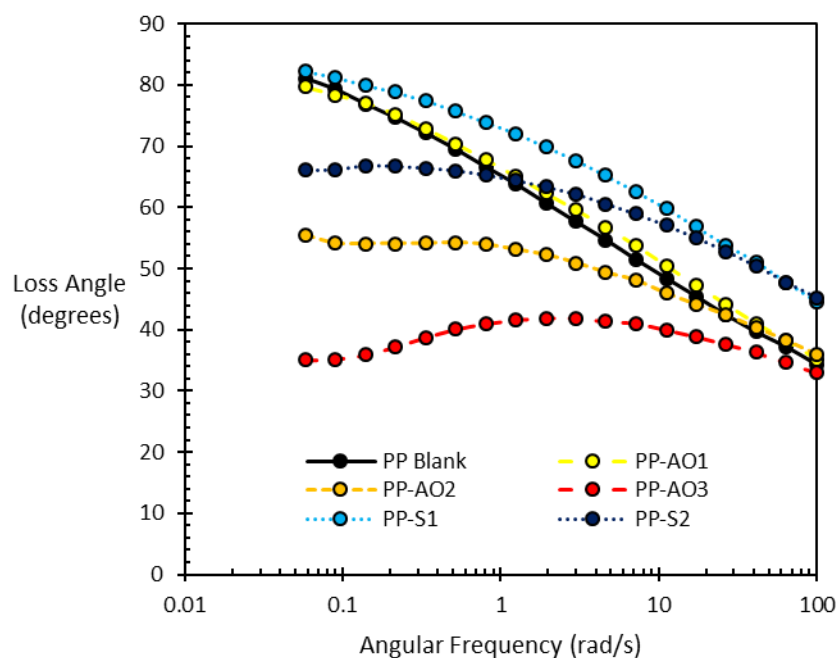


Figure 6 – Loss angles of branched PP samples

3.2.4 Extensional Rheometry

Even a small population of LCB material can have a remarkable impact on extensional rheology properties [4], producing a response known as strain hardening. This characteristic is needed for a polymer to perform adequately in manufacturing processes such as film blowing,

thermoforming, and foaming [2]. Strain hardening occurs when a positive deviation from the linear viscoelastic curve is observed with increasing strain. This is assessed by superimposing measurements of the time-dependent extensional viscosity of samples with a calculated linear viscoelastic curve derived from the Trouton relationship, in which the time-dependent extensional viscosity is three times the time-dependent shear viscosity [50].

Despite not exhibiting strong branching effects in shear rheology, PP-S1 exhibited strain hardening behaviour (Figure 7), a testament to extensional rheology's greater sensitivity to branching [4]. PP-S2 demonstrated very strong strain hardening, as anticipated from the result of the gel fraction analysis.

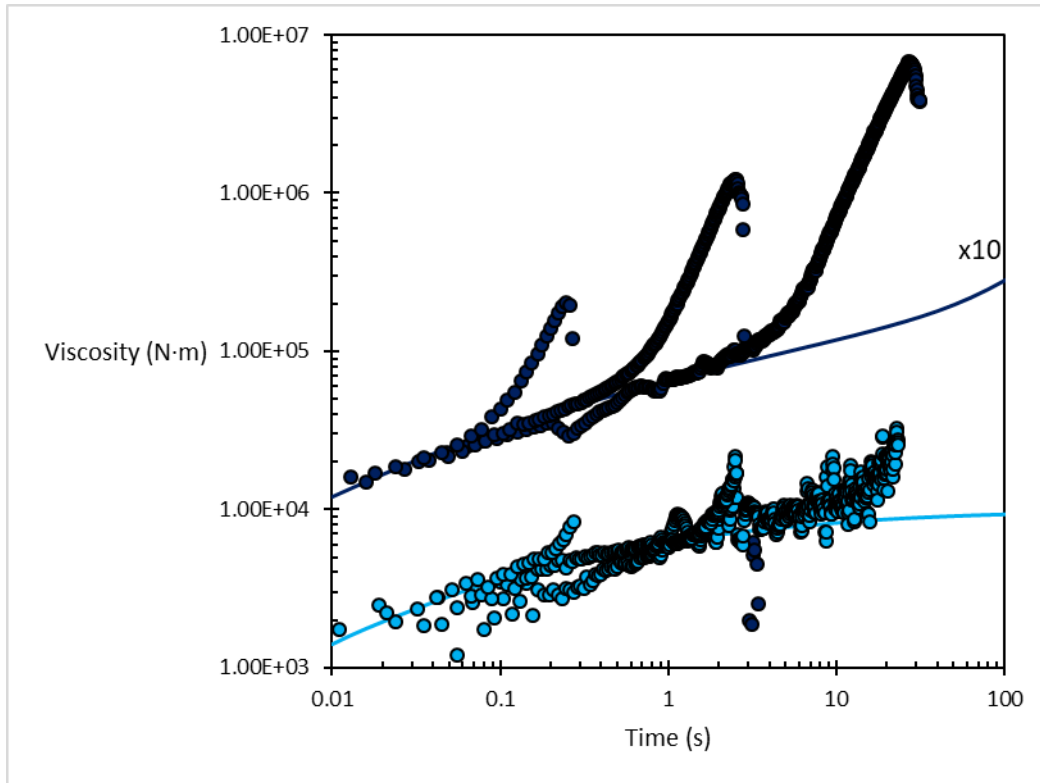


Figure 7 - Extensional rheology of silane-branched samples: • PP-S1, • PP-S2, solid lines are $3\eta_0^+$ (Trouton relation)

Extensional rheology measurements of the PP-*b*-AOTEMPO derivatives showed a variety of behaviours (Figure 8). Of the AOTEMPO-branched samples, both the gelled PP-AO3 and the gel-free PP-AO2 exhibit the strain hardening characteristic of long-chain branching [51]. While PP-AO1 shows slight deviation from the Trouton relation, this was only observed at low strain rates and is attributable to sagging effects [52]. As anticipated, higher loadings of the branching agent yield increased branching and in turn increased strain hardening [4].

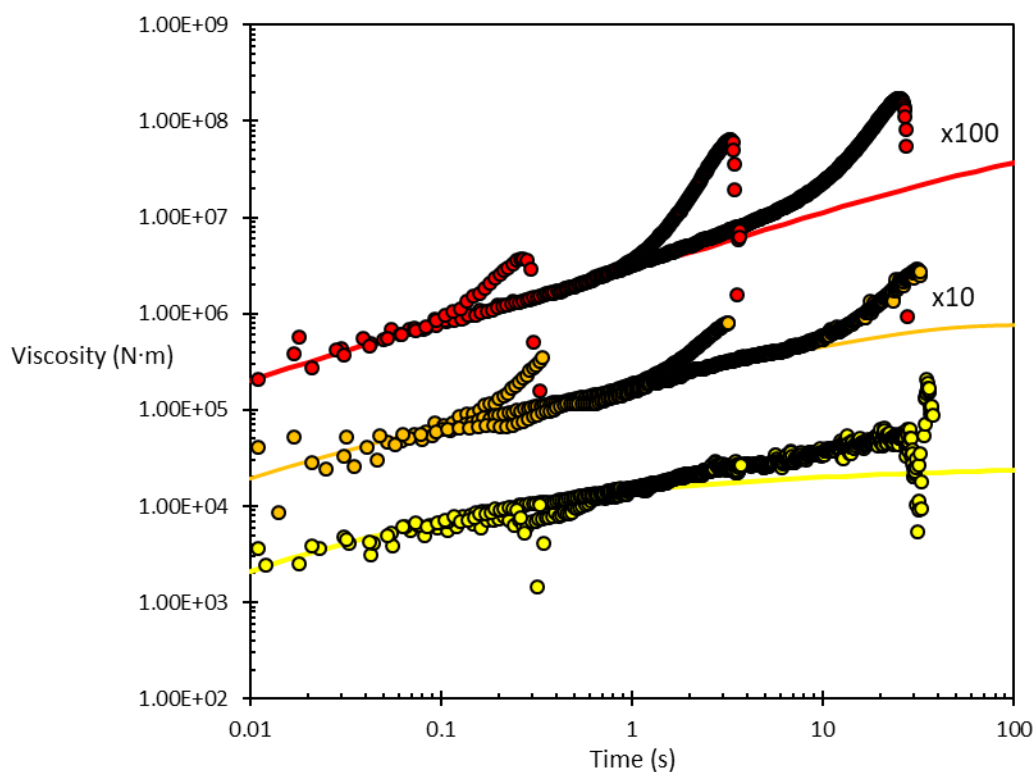


Figure 8 - Extensional rheology of AOTEMPO-branched samples:

• PP-AO1, • PP-AO2, • PP-AO3, solid lines are $3\eta_0^+$

While strain hardening can be identified visually from charts of extensional viscosity over time or strain, quantification assists in comparing the strain hardening behaviour between samples. One metric described by *Stange et al* [53] is the ‘strain hardening coefficient’ S , defined by

$$S = \frac{\eta_E^+(t, \dot{\epsilon}_0)}{3 \cdot \eta_0^+(t)},$$

where the denominator is the time-dependent elongational viscosity as derived from the Trouton relation, while the numerator is the measured elongational viscosity. As the Trouton relation will predict the behaviour of a linear polymer, S is the factor by which the extensional viscosity is increased as a result of a melt's nonlinear architecture. In general, decreasing strain hardening with increasing strain rate is expected of materials with a small number of branches of high molecular weight [50] [53].

The architecture of these branched species may be inferred from the means, method, and magnitude of the branching reaction. Using the abstraction efficiency from PP by DCP (37%) [7], PP-AO2 would be expected to contain an average of 0.88 grafts/molecule, of which not all will may successfully oligomerize to form branches. While the actual abstraction efficiency for L130 may prove slightly higher (yielding slightly higher grafts per chain), it should be considered that as H-abstraction is random and does not selectively occur to ungrafted chains, the polymer will contain chains possessing a varying number of grafts. As with other graft modifications, longer chains will receive more grafts. As the oligomeric chain linking PP chains is short, singly functionalized PP will yield star branch architectures, with the number of appendages determined by the kinetic chain length of the oligomerization. Multiply functionalized chains may produce pom-pom structures, or more complex branching architectures. Figure 9 shows two a many possible structures: the 'star' architecture would result from the oligomerization bonding two singly functionalized chains, while the 'pom-pom' shown requires that the oligomerization of

each grafted acrylate on a doubly functionalized chain incorporate two other singly-grafted chains.

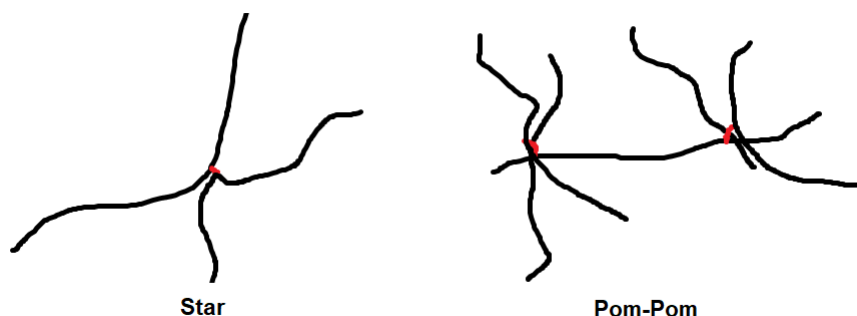


Figure 9 – Possible architectures of PP-b-AOTEMPO, depending on kinetic chain length and degree of functionalization

As average grafts per molecule are low, the star structure is likely more representative of the branched species produced in this work, though other branching structures are bound to be present but uncommon.

Gel-free LCB-PP samples that exhibited strain hardening, PP-S1 and PP-AO2, are of particular interest. As illustrated in Figure 9, strain hardening coefficients at a Hencky strain of 2 (lying above the onset of strain hardening, but below the ultimate strain for all trials) are similar for both materials. Additionally, the relationship between S and strain rate is related to the branch architecture, with the observed decline in S with increasing strain rate being consistent with the presence of few branches of high molecular weight [53].

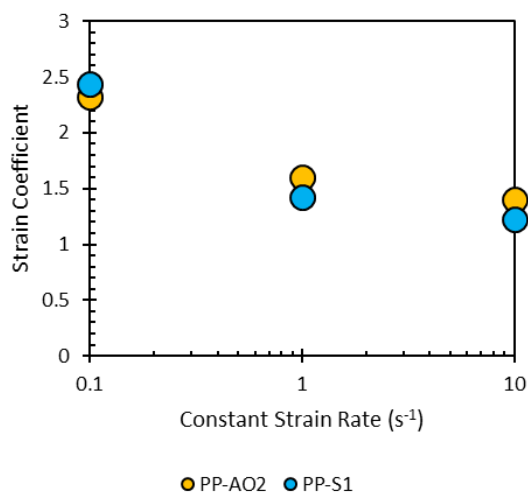


Figure 10 - Strain hardening coefficients for PP-S1 and PP-AO2 at a Hencky strain of 2

Although PP-S2 and PP-AO2 generated similar S values, the extensional viscosity of the AOTEMPO derivative is significantly greater. At a Hencky strain of 1 (below the strain hardening threshold) and a strain rate of $1 s^{-1}$, the extensional viscosity of PP-AO2 is 18000 Pa·s, compared to 5800 Pa·s for PP-S1. As with shear viscosity, this superior extensional viscosity is due to the reduced β -scission incurred by functionalized nitroxyl branching.

In addition to the inhibition of β -scission, AOTEMPO formulations are expected to offer more uniform branch distribution than existing methods. While not examined in this study, multifunctional monomer coagent-based grafting is known to produce bimodal weight and branching distributions because they generate hyper-branched nanoparticles in conjunction with a degraded material [10]. This architecture results from grafting, branching, and β -scission reactions all occurring simultaneously. While vinylsilane grafting decouples these reactions by replacing the radical crosslinking mechanism with a moisture-based one, an intramolecular abstraction mechanism results in a predilection for multiple functionalization of the same polymer chain [14]. Combined with the increasing difficulty of functionalizing ever-shorter chains

produced by β -scission, the effects on graft distribution are so severe that regardless of loading, silane-based crosslinking is only capable of achieving a gel fraction of around 80% in PP. In nitroxyl-based grafting, the main determining factor in graft distribution is macroradical distribution, with no further radical transfer reactions to dictate graft placement.

3.2.5 Crystallization Behaviour

LCB-PP crystallization is of interest due to the effects of crystallite density and size on solid-state material properties [54]. Previous investigations have found that branched PP samples crystallize from the melt state at high temperatures (T_c) than linear materials, but T_c increases are relatively insensitive to the degree of branching [55] [56]. In order to assess whether PP-*b*-AOTEMPO exhibited similar crystallization behaviour, nonisothermal DSC was performed at a scan rate of 5°C/min to produce the crystallization exotherms shown in Figure 11. Irrespective of the degree of PP functionalization and the chemistry employed, the T_c of each LCB-PP was 10-14°C greater than the parent material. Silane-branched samples and the gelled AOTEMPO-branched sample saw a slight increase in crystallinity, but changes were small and are attributable

to the sensitivity of the integration method. Different studies have shown both increased [54] [57] and decreased [58] [59] crystallinity in branched polyolefins.

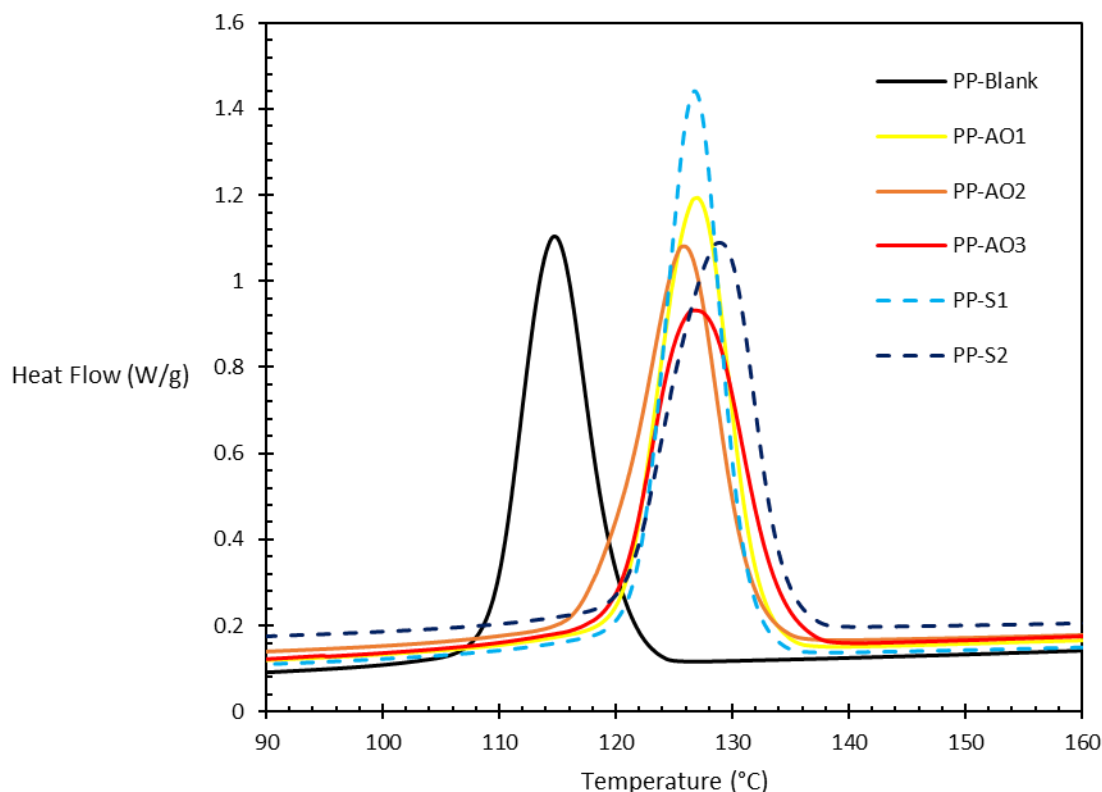


Figure 11 - Crystallization exotherms

Isothermal DSC data were acquired to probe the kinetics of LCB-PP crystallization. Due to the difference in T_c 's between the untreated PP and the branched derivatives, measurements were performed at both 130 and 140°C. These measurements revealed a stark differences between the branched materials and the starting polymer (Figure 12). Crystallization of LCB-PP derivatives at 130°C was very fast, reaching the ultimate degree of crystallization within 5 min, compared to over 35 min for the linear precursor. Raising crystallization temperature to 140°C

slowed the crystalline phase development for all PP materials, further increasing the deviation between LCB-PP and PP, while differentiating the lightly branched silane sample PP-S1 from the remainder of the field. Similar increases in isothermal crystallization rate were observed by *Zhang et. al.* [58], which silane-branched PP inducing a lesser reduction in crystallization time than coagent-based LCB materials.

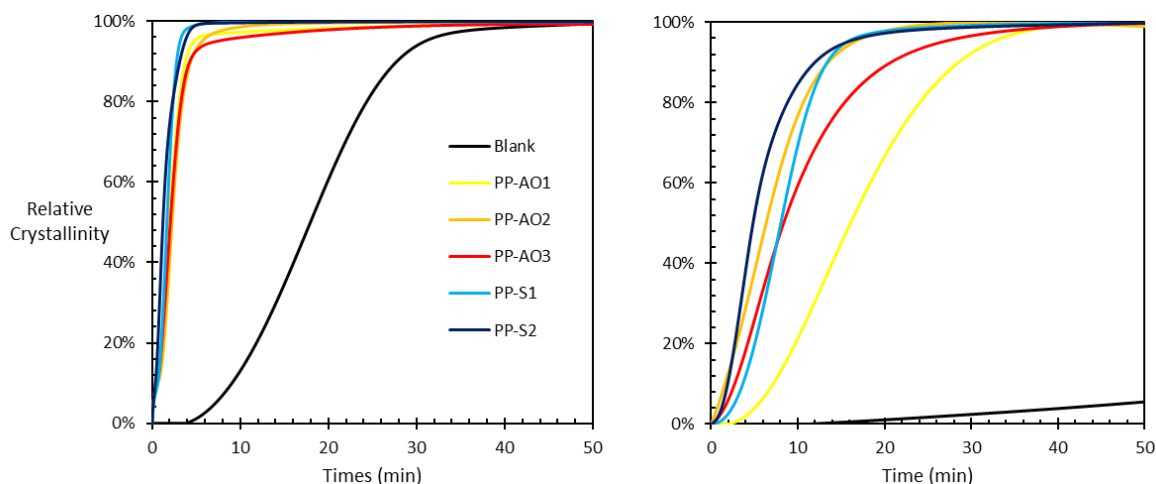


Figure 12 - Isothermal crystallization curves at 130°C (left) and 140°C (right)

Polarized optical microscopy measurements provide information on the evolution of semi-crystalline polymer morphology throughout the crystallization process and, as such, complement DSC data [5] [54] [55] [56]. These images illustrate the genesis and growth of spherulites in the PP melt, with the ultimate morphology depending on the rate of crystallite nucleation in relation to the rate of spherulite growth. The images presented in Figure 13 show that relative to the linear PP “blank” crystallization process, the gel-free AOTEMPO derivative PP-AO1 nucleates spherulites much more rapidly, resulting in accelerated crystallization and smaller final spherulite size. This is consistent with recent reports on PP-LCB crystallization, which attributed this behaviour to a nucleating effect of branch points [5]. Although definitive

conclusions regarding the nucleation mechanism cannot be drawn from available information, the prodigious nucleation observed in these gel-free branched samples supports the hypothesis that branched structures contribute in some manner to the nucleation of polypropylene crystals.

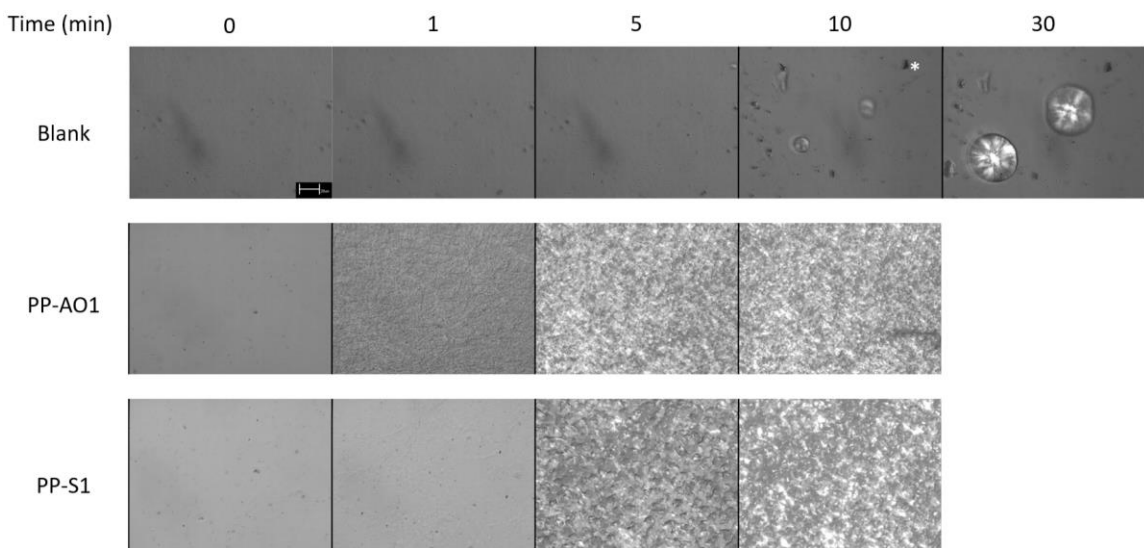


Figure 13 - Polar optical microscopy of polymer crystallization at 140°C. Scale bar is 20 μm .

3.3 Conclusion

Peroxide-initiated formulations containing AOTEMPO were used in a single-stage, melt-state modification of linear PP to yield gel-free LCB derivatives without incurring degradation of the polymer matrix. These high viscosity PP-LCB materials provide the extensional strain hardening characteristics required by key polymer processing operations and cannot be otherwise manufactured by conventional technology. This accomplishment is attributable to this chemistry's inhibition of the β -scission normally incurred in radical-mediated modification, which preserves the integrity of the polymer matrix. Additionally, the AOTEMPO approach is expected to yield even uniform branching and molecular weight distributions than existing LCB chemistry.

Chapter 4

Functional Nitroxyl Formulations for Polypropylene Graft Modification

4.1 Introduction

The radical grafting of maleic anhydride (MA) to polyolefins is commercialized technology that improves adhesion to high surface energy fillers and blend components [60]. Unfortunately, the functionalization of PP is accompanied by degradation of the polymer matrix, resulting in maleated products of very low melt viscosity [8]. While high molecular weights and high graft content can be accomplished using *N*-bromosuccinimide as an additive, this chemistry racemizes PP and reduces crystallinity [25].

Previous work has shown that functional nitroxyls can alter the architecture of PP without compromising molecular weight. Formulations containing acrylated TEMPO (AOTEMPO) [61] or vinylbenzylated TEMPO (VBTEMPO) [33] have been shown to crosslink PP to high gel contents, yielding thermoset derivatives that are inaccessible by conventional graft-modification chemistry due to β -scission. Nitroxyls bearing alternate functional groups have also been used to prepare high molecular weight PP derivatives bearing pendant alkoxyamines by the nitroxyl coupling reaction (NCR) [27]

In addition to being used with peroxide alone, both AOTEMPO and VBTEMPO have been used in combination with multifunctional monomers possessing compatible oligomerizable functional groups. In the case of AOTEMPO, the effectiveness of the crosslinking agent trimethylolpropane triacrylate (TMPTA) was increased drastically when used in conjunction with the functional nitroxyl [61], producing crosslink yields far exceeding those accessible with the coagent alone. The synergy between functional nitroxyls and coagent is attributed to the efficiency of macromonomer production by NCR, which means that TMPTA does not need to

graft to the polymer by C-H bond addition, but only through copolymerization with polymer-bound acrylate functionality.

This is not to say that NCR is a highly efficient functionalization process. The alkoxyamine yield is dictated by the amount of peroxide in the formulation and the rate of H-atom abstraction from PP compared to the rate of alkoxy radical fragmentation to methyl radical and ketone. The H-atom abstraction efficiency is about 35% for cumyloxyl radicals acting on PP, meaning that 65% of the alkoxy radicals generated by the peroxide produce methyl radicals that will be trapped irreversibly by the nitroxyl [7] [29]. Moreover, unlike radical grafting by C-H bond addition to an olefin, which can produce multiple grafts using a single radical, NRC has no kinetic chain length. Each peroxide-produced macroradical can only yield a single graft. Fortunately, the resulting alkoxyamine bears functionality that reacts with a very good kinetic chain, thereby leveraging the impact of each nitroxyl coupling event.

It is known that maleic anhydride (MA) and maleimides will not homopolymerize readily under PP melt modification conditions [22], but will undergo alternating radical copolymerization with electron-rich monomers such as styrene and vinyl ethers [37]. Previous work has used styrene as a comonomer in MA grafting to increase the graft yield achieved for a given extent of PP degradation [26].

In the present work, a vinyl ether-functionalized nitroxyl (VETEMPO) is used to provide an MA copolymerization pathway, yielding a vinyl ether-*alt*-maleic anhydride copolymer that is bound to the polymer matrix. Note that this oligomerization may also engage methyl alkoxyamines to improve polymer-bound MA content.

4.2 Results and Discussion

4.2.1 Model Synthesis of BVE-*alt*-MA

Alternating copolymers of maleic anhydride and vinyl ethers are generally synthesized in the bulk or in concentrated solution to improve reaction yields and molecular weight. Some studies have explored similar polymerizations at low monomer concentrations: *Tsujii et al* [62] prepared thermoset copolymers using 2,4-dimethyl-1,3-pentadiene and maleic anhydride at concentrations of 0.5 mol/L in chloroform, while *Wang et al* [39] prepared the copolymer of maleic anhydride and isobutyl vinyl ether at initial concentrations of 0.2 and 0.4 mol/L (respectively) in a 3:7 solution of cyclohexane and ethyl acetate. Additionally, these copolymerizations were performed at temperatures below 100 °C, generally in the range of 30-80 °C. This is significantly below the temperatures required for solvent-free PP modification, which must be conducted above the ~165 °C melting point of the polymer.

Based on existing literature, the feasibility of accomplishing the maleic anhydride-vinyl ether copolymerization in a polypropylene melt was uncertain, owing to the nonpolar character of the polymer melt and the process temperature. Therefore, a model synthesis of n-butyl vinyl ether-*alt*-maleic anhydride was conducted at 180 °C in tetradecane, with the hydrocarbon taking the place of PP.

Upon heating, the reaction solution turned cloudy due to the formation of a precipitate, indicating the synthesis of the copolymer. The IR spectrum of the copolymer (Figure 14) shows symmetric and antisymmetric stretches of the MA carbonyl groups at 1780 and 1864 cm⁻¹, respectively, as well as the stretching of the ether linkage at 1225 cm⁻¹ [63].

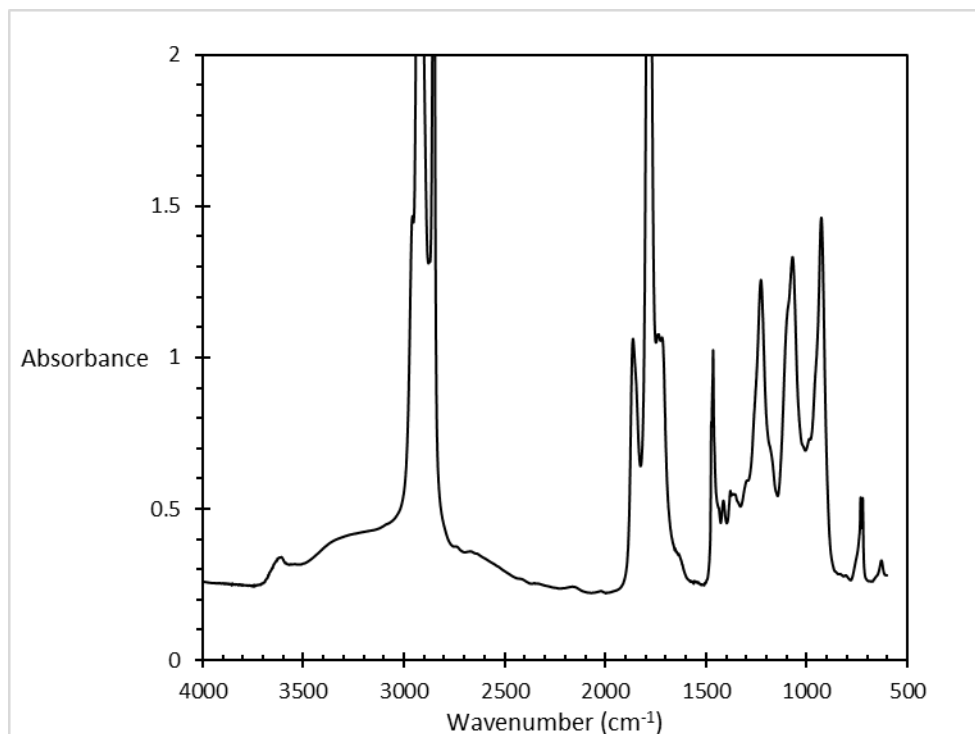


Figure 14 - IR spectrum of BVE-*alt*-MA

A strictly alternating copolymer will have an equimolar composition of the constituent monomers. The equimolar composition of BVE-*alt*-MA was confirmed by ^1H spectroscopy. Reacting the copolymer with benzylamine converted the anhydride to the corresponding amidic acid, facilitating NMR characterization by introducing downfield aromatic resonances. Integration of the five aromatic protons and the three methyl protons of the butyl ether yielded a 5:3 ratio, which is consistent with an equimolar composition.

Given the equimolar composition of the product and literature reports of the alternating nature of vinyl ether/maleic anhydride copolymerizations [38] [63], it is reasonable to conclude that the polymer prepared in this study is alternating. More importantly, this demonstration confirms that maleic anhydride and vinyl ethers will copolymerize within a non-polar hydrocarbon matrix, even at low initial monomer concentrations (<0.08 mol/L). As such, the

polymer-bound alkoxyamine bearing vinyl ether that is introduced by VETEMPO is expected to copolymerize with maleic anhydride under typical PP melt reaction conditions.

4.2.2 Anhydride Grafting with VETEMPO

The anhydride content of graft-modified PP can be quantified by absorbance integration of IR spectra or by back titration of the hydrolyzed anhydride functionality. Due to procedural difficulties involved in titration [64], the IR technique was adopted wherein the area of the carbonyl stretching absorbance is divided by the absorbance of a reference peak associated with the base resin to calculate a carbonyl index (CI):

$$CI = \frac{A_{1760-1812}}{A_{2695-2853}}$$

The millimoles of anhydride functionality per gram of polymer was derived from CI measurements using calibrations derived from mixtures of dodecenyl succinic anhydride (DDSA) and BE-*alt*-MA for PP samples maleated using VETEMPO. Composition-specific calibrations were required because structural neighbours affect the molar absorptivity of carbonyl resonances. Previous studies involving styrene as a maleation comonomer required a separate calibration curve for the MA-styrene copolymer [65]. The calibration curves presented in Figure 15 show that absorptivity of the alternating copolymer is just two-thirds that of the succinyl group.

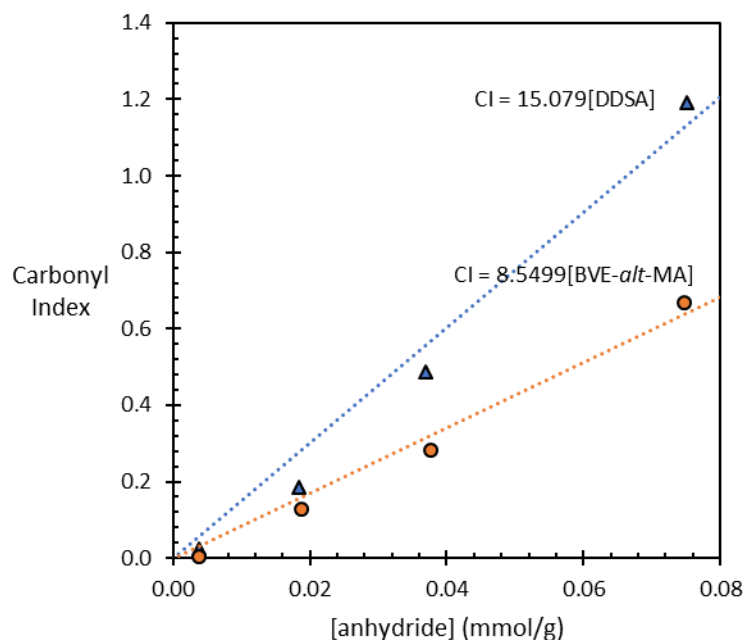
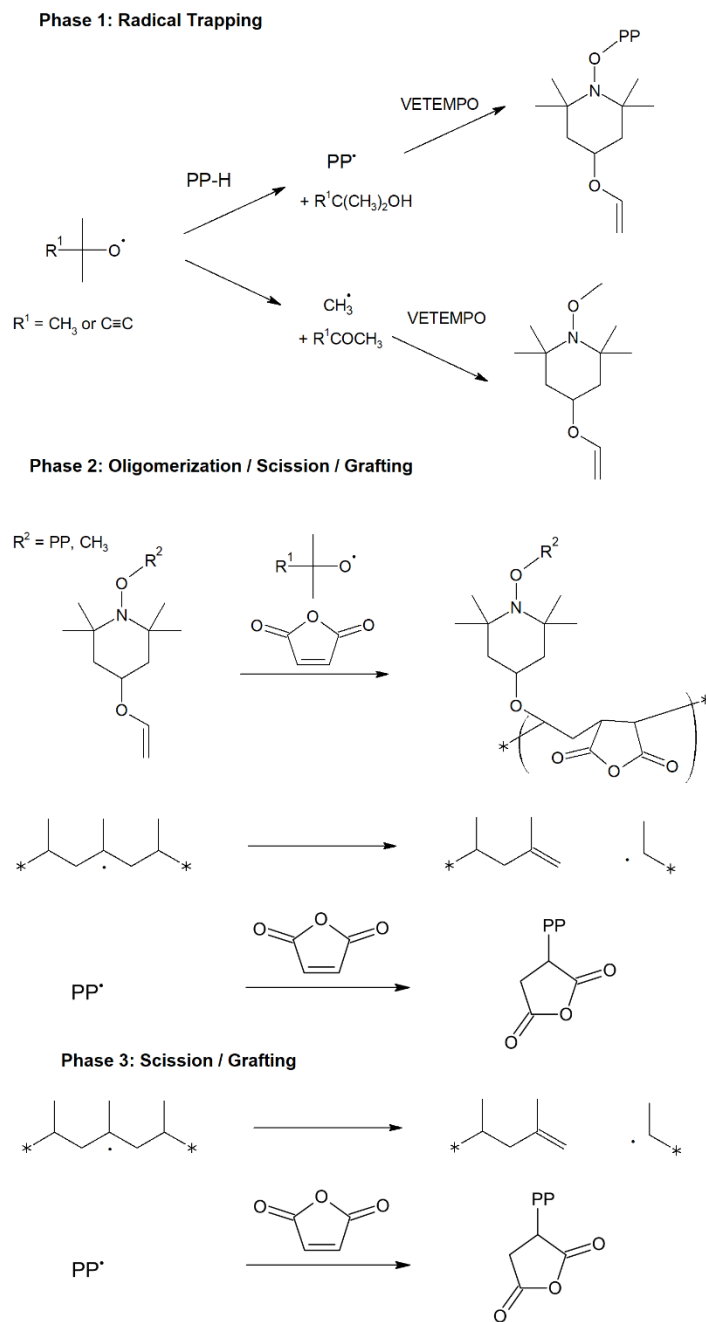


Figure 15 - Calibration curves prepared for PP-*g*-MA and PP-*g*-VETEMPO/MA

This difference in IR sensitivity complicates attempts to quantify the functional group concentration within PP-*g*-VETEMPO/MA. Scheme 2 illustrates a simplified mechanism for radical-mediated modification involving VETEMPO and maleic anhydride. As with previous functional nitroxyl modifications of PP [33] [61], other radical-mediated reactions proceed concurrently with the oligomerization of the nitroxyl-grafted functionality (radical trapping, due to its speed, inhibits other radical activity). Because co-oligomerization of the vinyl ether with MA will occur alongside radical addition of maleic anhydride, PP-*g*-VETEMPO/MA will contain both MA incorporated through oligomerization and singly grafted MA. Therefore, the composition of PP-*g*-VETEMPO/MA derivatives will be presented as a range spanning from a lower limit (assuming 100% single graft content) to an upper limit (assuming 100% copolymer content).



Scheme 2 - Simplified reaction mechanism of VETEMPO-assisted grafting

The main drawback of conventional PP maleation technology is polymer degradation, with increased grafting content gained at the expense of melt viscosity. Therefore, the performance of alternative methods should be evaluated on selectivity of maleation versus

degradation in comparison to that provided by standard peroxide/MA formulations. To this end, the effect of VETEMPO at trapping ratios of 0.5, 0.75, and 1 was tested against PP-*g*-MA samples prepared at different peroxide loadings, as listed in Table 2.

Table 2 - Recipes used to prepare maleated PP

Sample	[L130] ($\mu\text{mol/g}_{\text{PP}}$)	[MA] (wt%)	[VETEMPO] ($\mu\text{mol/g}_{\text{PP}}$)	Graft degree (wt%)	Graft efficiency (%)
PP-0	0	2	0	0.00	0
PP-2	2.0	2	0	0.28	14
PP-4	3.9	2	0	0.50	25
PP-10	9.8	2	0	0.67	33
PP-TR0.5	3.9	2	7.8	0.41–0.72	20–36
PP-TR0.75	3.9	2	11.7	0.43–0.75	21–38
PP-TR1	3.9	2	15.6	0.22–0.39	11–19

Since co-oligomerization of vinyl ether functionality and MA can introduce long-chain branching, melt-state shear rheology acquired at low oscillation frequencies may show higher melt viscosities compared to linear PP of similar molecular weight [4]. At higher shear frequencies, long-chain branching (LCB) has little influence on viscosity, with molecular weight being the main determinant.

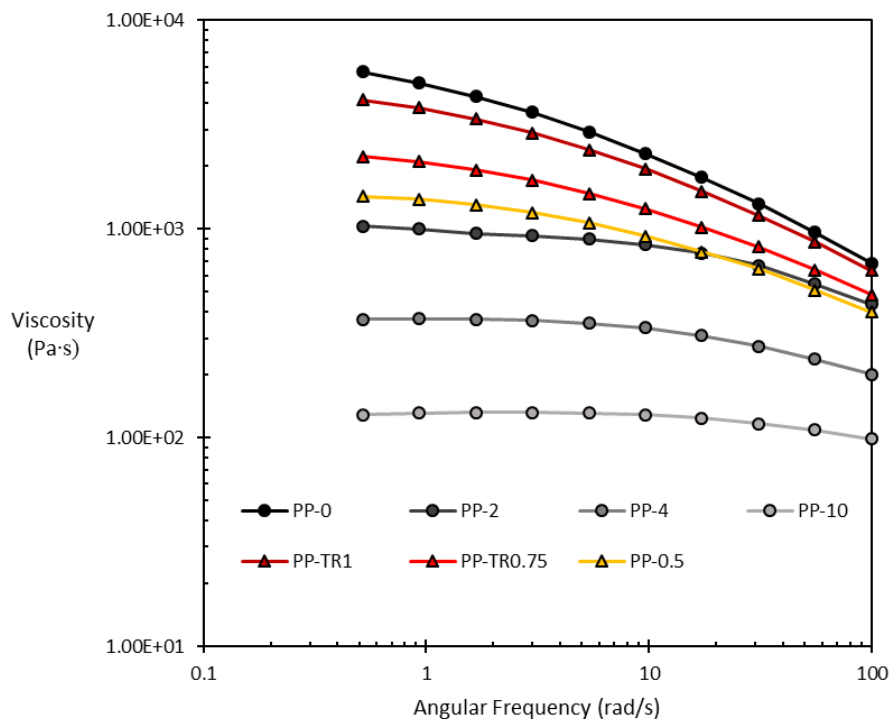


Figure 16 - Shear rheology of maleated polypropylene derivatives

The data presented in Figure 16 confirm that VETEMPO applied at trapping ratios of 0.5 and 0.75 generate LCB in maleated PP-derivatives. Although PP-2 (a linear material) and PP-TR0.5 and PP-TR0.75 exhibit similar complex viscosities at 100 rad/s, the maleated materials produce significantly higher viscosities at 1 rad/s. While these branching effects are less dramatic than those reported in **Error! Reference source not found.**, they are indicative of the co-oligomerization mechanism enabled by VETEMPO.

Comparing the graft content and high-frequency viscosities (where LCB has little effect) of PP-g-VETEMPO/MA to those of standard maleation products reveals remarkable improvements (Figure 17). For example, a conventional formulation of 3.9 $\mu\text{mol/g}_{\text{PP}}$ L130 and 2 wt% MA gave a product containing 0.5 wt% anhydride functionality with a complex viscosity of 201 Pa·s at 100 rad/s. Charging VETEMPO to this formulation at a trapping ratio of 0.75 gave a

product that contained at least as much anhydride functionality with a complex viscosity of 484 Pa·s at 100 rad/s.

Lowering the VETEMPO trapping ratio from 0.75 to 0.5 gave in a lower viscosity product. This is presumably due to the actions of a more extensive phase 3 (Scheme 2), wherein complete vinyl ether consumption leaves PP macroradical scission as the sole molecular weight altering process. Surprisingly, increasing the trapping ratio from 0.75 to 1.0 produced an appreciable anhydride graft content. This nitroxyl concentration is sufficient to quench all initiator-derived radicals, leaving no opportunity for any polymer modification beyond the introduction of alkoxyamine functionality. Acknowledging that a slight decrease in viscosity can be suffered due to thermal degradation (trapping ratios of 1 using AOTEMPO yielded similar losses, as shown in **Error! Reference source not found.**, Figure 3), the introduction of anhydride functionality to the polymer is unusual. The origins of this functionalization are uncertain, but

several hypotheses present themselves.

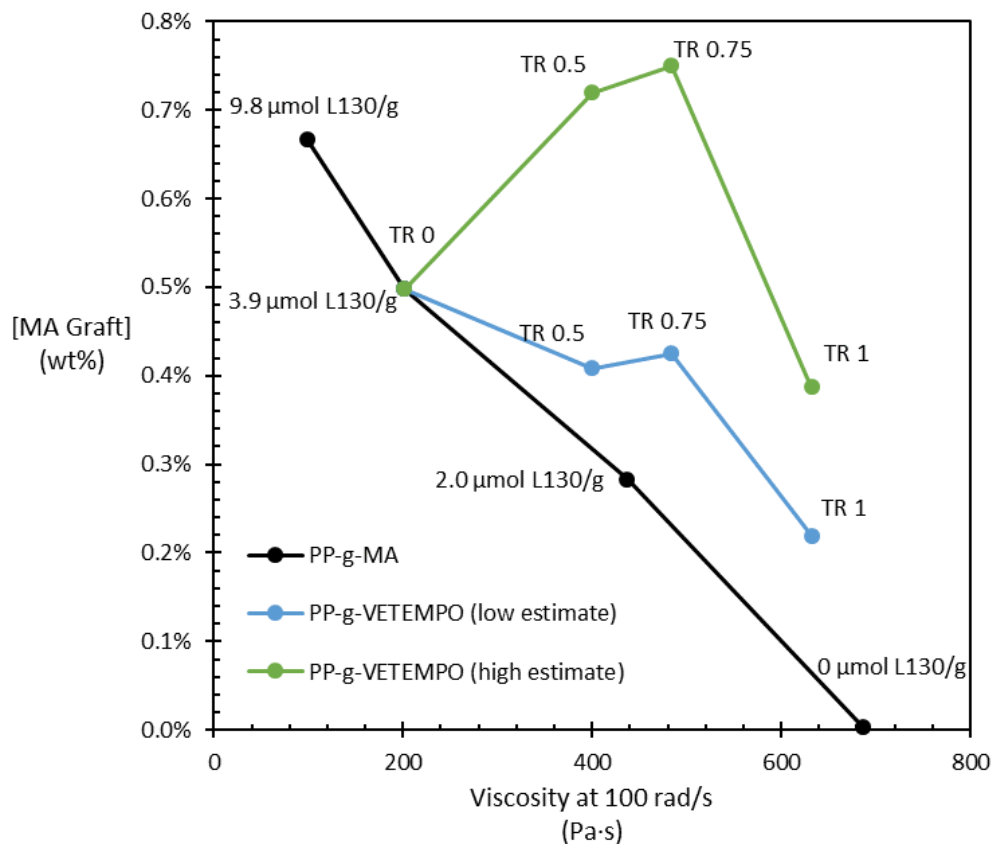


Figure 17 - Viscosity and graft content of PP-g-VETEMPO/MA relative to PP-g-MA. Low estimate assumes no copolymer anhydride incorporation, high estimate assumes all incorporated anhydride is in the form of the copolymer

While nitroxyl trapping is several orders of magnitude faster than other radical addition reactions [29], reaction rate is nevertheless proportional to concentration. As the nitroxyl concentration decreases, it may be possible for some oligomerization to occur, incorporating significant quantities of the anhydride due to the high vinyl ether loading. Alternately, some charge-transfer complexes are known to undergo spontaneous polymerization [36]. Provided that the vinyl ether was successfully grafted to the polymer, a spontaneous initiation by the vinyl ether

and the anhydride could account for the oligomerization taking place despite the lack of available peroxide-produced radicals. However, this cannot be the case given the behaviour of a trapping ratio 1 system observed later in this chapter. A further possible explanation relies on the thermal instability of the tertiary alkoxyamine [33] that is formed in significant quantities by the nitroxyl trapping of macroradicals. It has been previously demonstrated that another unstable alkoxyamine, formed by trapping polybutadiene macroradicals with TEMPO, allowed for the insertion of maleic anhydride as unsaturated grafts [66].

Regardless the unusual behaviour of systems at trapping ratios of 1, VETEMPO at a trapping ratio of 0.75 yields MA-functionalized PP derivatives with a unique combination of high graft content and high viscosity that is unattainable through conventional maleation technology.

4.2.3 PP Thermoset Production using Multifunctional Monomers

Simple peroxide-based methods for crosslinking linear PP homopolymers beyond the gel point remain an area of interest in the polyolefin modification field, since these thermoset derivatives can improve high temperature service performance. Functional nitroxyls bearing acrylate and vinylbenzoate functionality have proven effective in transforming peroxide-only formulations from degradative to molecular weight building. However, achieving very high crosslink densities generally required high peroxide and nitroxyl loadings in combination with a highly reactive functionality [33]. An alternate approach involving VETEMPO acting in concert with a bis-maleimide coagent, 1,3-phenylene dimaleimide (PDM), was assessed as a potential crosslinking process that exploits the high co-oligomerization activity between vinyl ethers and electron-deficient monomers such as N-aryl maleimides.

Baseline performance markers were acquired for PP modifications involving TEMPO, AOTEMPO and VETEMPO in conjunction with maleic anhydride. The data plotted in Figure 18 are time-resolved, melt-state rheology measurements of dynamic storage modulus (G'), which

serves as a proxy for crosslink density [67]. In the absence of nitroxyl, PP degradation was rapid and extensive, as evidenced by losses in G' . However, the inclusion of any nitroxyl produced the expected an induction period that results from macroradical quenching [34]. Following this induction period, the TEMPO formulation exhibited significant degradation as PP scission proceeded unimpeded. Formulations containing AOTEMPO or VETEMPO also suffered degradation in the post-trapping phase, presumably due to thermal degradation as well as the instability of tertiary alkoxyamine. These results are consistent with those presented in section 4.2.2, in that the co-oligomerization of vinyl ether and MA at the reagent loadings used in this work can introduce LCB to a PP derivative, but cannot crosslink it to the point of thermoset production.

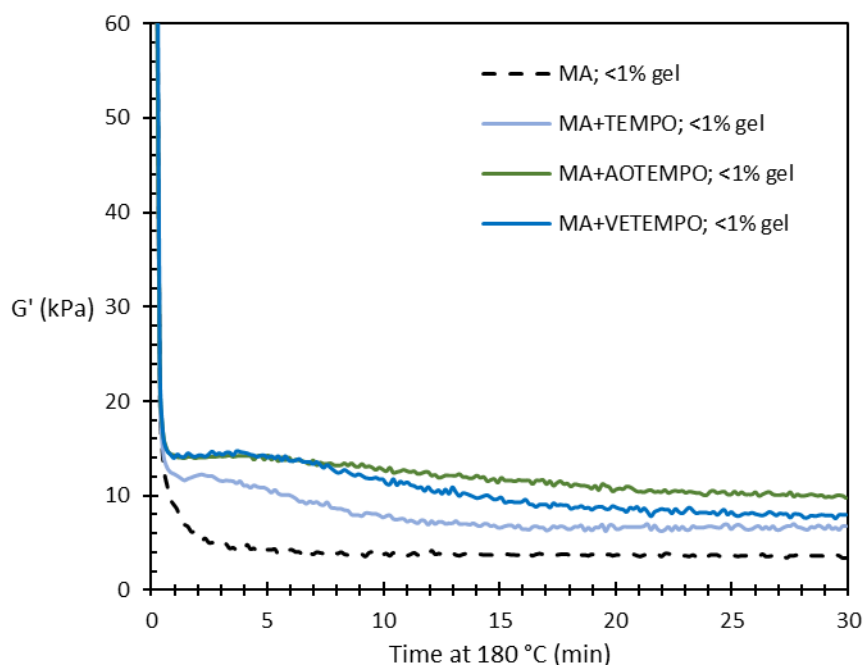


Figure 18 - Influence of functional nitroxyl on PP cure dynamics ([MA] = 2 wt%, TR = 0.75, [L130] = 9.8 $\mu\text{mol/g}$)

More efficient crosslinking was accomplished by replacing MA with the difunctional monomer PDM, as evidenced by the storage modulus data presented in Figure 19. Starting from an initial value of 15 kPa, a combination of PDM and VETEMPO at a trapping ratio of 0.75 generated an ultimate storage modulus of 40 kPa and a 30% gel content within 30 min at 180 °C. This compares favourably to PDM, which produced a reduced-modulus product with a gel content of only 18%. As expected, the monofunctional N-phenyl maleimide (NPM) yielded similar reaction dynamics to the MA system, demonstrating the efficacy of PDM is due to this difunctionality as opposed to its maleimide structure (in contrast to the anhydride).

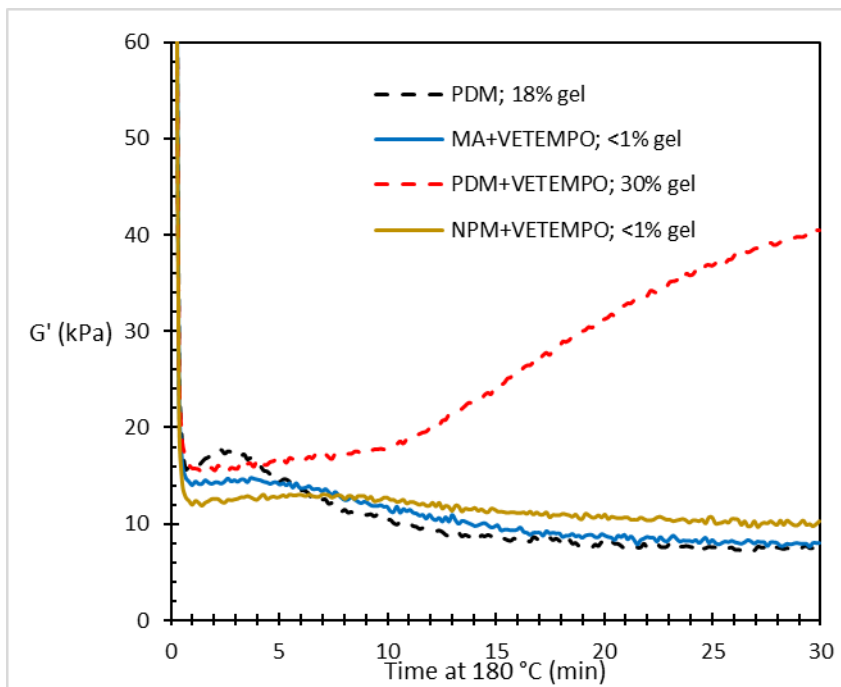


Figure 19 – Influence of monomer on PP cure dynamics ([monomer] = 2 wt%, TR = 0.75, [L130] = 9.8 μ mol/g)

The 0.75 trapping ratio used in the experiments summarized in Figure 19 was based on earlier work on polypropylene thermosets synthesis [33]. However, the modulus continued to rise after 30 min, suggesting that the system remained in phase 2 (Scheme 2) without progressing to a

phase 3 cure reversion. A more expansive study of VETEMPO trapping ratio effects is presented in Figure 20. Lowering the trapping ratio from 0.75 to 0.5 gave a higher ultimate G' and gel content, despite producing less polymer-bound vinyl ether. Evidently, what vinyl ether functionality was produced was converted to crosslinks more completely, owing to the greater availability of peroxide-derived radicals to initiate the oligomerization process. Dropping the trapping ratio further to a value of 0.25 resulting in a significant reversion of the storage modulus, but not significant effect on gel content.

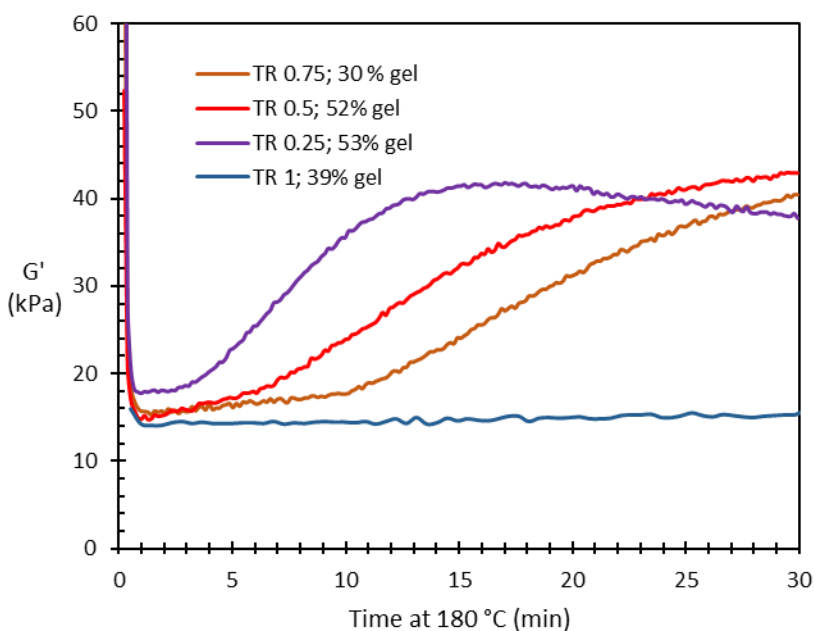


Figure 20 – Influence of VETEMPO trapping ratio on PP cure dynamics ([PDM] = 2wt%, [L130] = 9.8 $\mu\text{mol/g}$)

Raising the trapping ratio to 1.0 suppressed all crosslinking/degradation activity, given a dearth of peroxide-derived radicals to support co-oligomerization. This stable modulus was observed 90 minutes, which shows that a spontaneous self-initiation of the radical oligomerization of vinyl ether and maleimide is not significant.

While certainly a less powerful crosslinking agent than the previously studied VBTEMPO, VETEMPO provides a novel capability as a crosslinking agent. The crosslinking reaction of VETEMPO necessarily involves a polar monomer, either maleic anhydride or a maleimide. This provides a single process by which both polar functionalization and crosslinking can be accomplished. In contrast, a maleated PP thermoset prepared using another functional nitroxyl would require separate means of grafting the anhydride to the polymer and crosslinking the chains. The two mechanisms would be in competition for radicals. While maleated polypropylene can be crosslinked without further radical chemistry using multifunctional monomers reactive with the anhydride [12], conventionally-produced PP-*g*-MA's low molecular weight makes this a questionable synthetic route toward PP thermosets.

4.3 Conclusion

The ability of the vinyl ether functionality to copolymerize with maleic anhydride and maleimides presents new possibilities for functional nitroxyl modification of polymers. By incorporating the monomer through oligomerization rather than radical addition, β -scission is reduced, and higher molecular weight derivatives can be produced.

Without a multifunctional coagent, functional nitroxyls are only suitable for producing densely crosslinked PP at high loadings and a highly reactive oligomerizable functionality must be chosen. The addition of a multifunctional monomer such as PDM greatly increases the crosslinking yield. This is particularly important given the relative novelty of functional nitroxyl chemistry, and their greater cost compared to these readily available monomers.

For scission-prone polymers such as PP, trapping ratio remains a key factor in accomplishing efficient modifications. An insufficient trapping ratio will allow cure reversion to undo the gains made by the oligomerization reaction, while excessive nitroxyl will not leave sufficient radicals to take advantage of the grafted functionality.

The results achieved with VETEMPO demonstrate the potential of functional nitroxyl chemistry to not only perform structural modification of scissionable polymers such as long-chain branching and crosslinking, but also incorporate other monomers. Here, VETEMPO was used to perform the industrially relevant maleation of PP with reduced scission. Beyond maleation, any number of modifications may be possible, provided the desired graft functionality can be paired with a suitably functionalized nitroxyl.

Chapter 5

Conclusions and Future Work

5.1 Conclusions

Prior to the present work, the use of oligomerizable functional nitroxyls focused on producing PP thermosets with controlled dynamics and yields. This investigation broadens the applications of functional nitroxyl chemistry to include long-chain branching and graft modification.

Alternating copolymer systems involving maleic anhydride (and maleimides) and vinyl ethers were shown to be polymerizable at low concentrations in an olefinic medium. This copolymerization was implemented as a novel means of MA functionalization of PP using VETEMPO. Use of a multifunctional monomer, PDM, enables the derivatization of PP thermosets using VETEMPO.

While traditional methods of radical modification of PP must contend with a powerful β -scission mechanism, functional nitroxyl chemistry circumvents this problem with rapid trapping of PP macroradicals, preventing scission. Through inhibition of chain degradation, this chemistry allows the production of a wide range of high-weight and crosslinked PP derivatives.

5.2 Future Work

5.2.1 Kinetics

Presently, understanding of the kinetics involved in functional nitroxyl modifications is quite limited. The dominant rate of the trapping reaction is well-understood, but the rate constants cited are based on small-molecule studies. The impact of melt viscosity on radical diffusion is of particular concern, given that oligomerization of the grafted groups relies on physical proximity. Particularly in scission-prone polymers, the relative rates of the oligomerization and scission reactions are also of concern.

5.2.2 Alkoxyamine Instability

While the instability of the tertiary alkoxyamine has been documented, its influence in the context of PP modification has not been explored. Given the unusual maleation activity observed when VETEMPO was applied at a trapping ratio of 1.0, the thermal instability of thermal alkoxyamines should not be a footnote in studies of nitroxyl chemistry involving PP starting materials.

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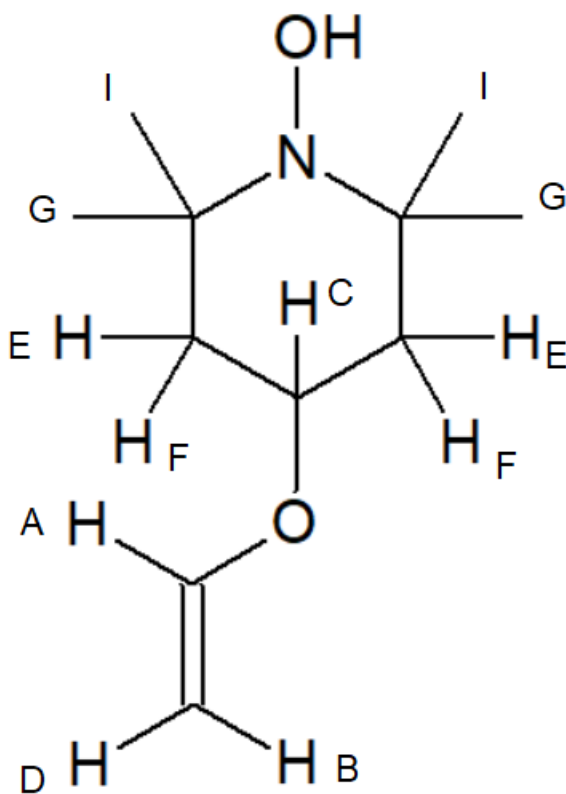
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Appendix A

^1H NMR Assignment of VETEMPO

In this work, VETEMPO was reduced to a diamagnetic derivative using phenylhydrazine. Scheme 3 shows the labelled molecular structure of the compound, while the spectrum itself is found in Figure 21. Peaks at 6.7 and 7.1 ppm are the phenylhydrazine and the oxidized derivative.



Scheme 3 - Molecular structure of reduced VETEMPO with proton labelling corresponding to the ^1H NMR spectrum in Figure 21

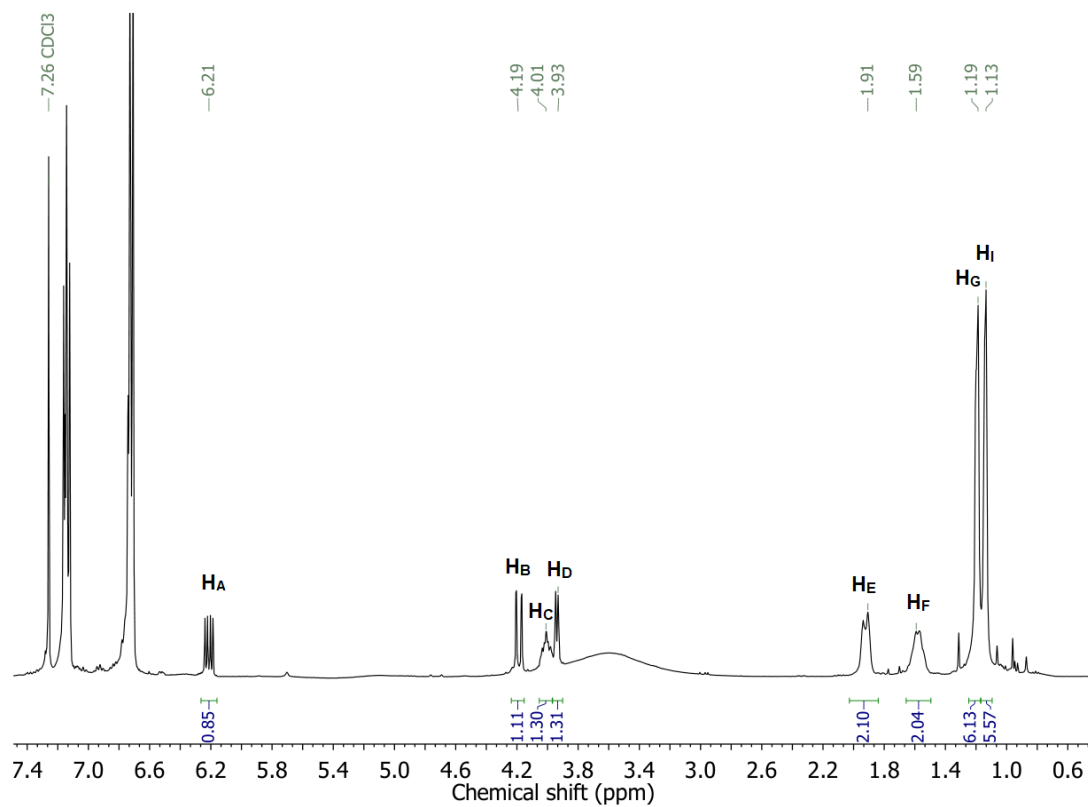


Figure 21 - ¹H NMR spectrum of VETEMPO reduced with phenylhydrazine. Solvent is CDCl₃